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4E4 Industrial Internship Project

In Partnership with ElectroRoute



ElectroRoute

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A dissertation submitted in partial fulfilment of the degree:
M.A.I - Master of Mechanical and Manufacturing Engineering

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Abstract

This report presents the work undertaken during a nine-month industrial internship with ElectroRoute, an international energy trading and services company. The internship was based within the Trading Operations team, which is responsible for the real-time management of physical energy positions and the optimisation of renewable assets across the Irish and GB electricity markets.

The first project involved the design and development of a web-based KPI dashboard to track trading desk performance. Built using Flask as the backend framework and Plotly for interactive chart generation, the dashboard provides a comprehensive view of portfolio performance across three asset groups — DAM, RFIT, and SOL — with functionality to filter by date range and aggregation frequency. Key metrics including net income, capacity factor, dispatch down rates, day-ahead balancing costs, trader performance, and market analytics are displayed through a series of interactive charts and summary widgets. The dashboard was presented to the CEO and wider team, and documentation was produced to enable ongoing maintenance after the internship.

The second project involved a detailed analysis of solar trading strategy, conducted in collaboration with a senior trader. Forecast accuracy was evaluated across multiple auction stages and forecast providers using a dynamic ramp-based filtering approach. Backtesting of alternative auction split strategies revealed that reallocating a portion of volume from the day-ahead market to later intraday auctions could improve revenue outcomes. Key drivers of solar book performance were identified, including systematic forecast over-prediction, curtailment exposure during peak solar hours, and the influence of wind generation on electricity prices. A framework for a predictive ensemble model to determine optimal daily auction splits was proposed.

The third project involved the development of a machine learning ensemble model to forecast system-wide electricity demand. Ridge regression was used to produce a weighted blend of two existing demand forecasts — one from EirGrid and one from an external provider — exploiting their complementary strengths. The blended model achieved a Mean Absolute Error (MAE) of 102.66 MW and a Root Mean Square Error (RMSE) of 134.07 MW on unseen test data, outperforming both individual forecasts across all key accuracy metrics. The near-identical training and test errors confirmed that the model generalised well and had not overfit to the training data.

In addition to these technical projects, a structured trading strategy framework was developed and applied through a paper trading exercise on the REFIT asset portfolio in the Day-Ahead Market. The framework codified the key factors influencing trading decisions — including weather conditions, system fundamentals, forecast confidence, interconnector flows, SNSP, and Net Imbalance Volume — into a set of guiding principles for position construction. Paper trading provided a daily feedback loop to iteratively refine this framework, developing practical intuition for how forecast uncertainty translates into financial risk and how to synthesise multiple conflicting market signals into a coherent trading position.

Across all projects and activities, the internship developed strong technical skills in Python, SQL, and data visualisation, as well as a deep understanding of renewable energy trading, electricity market structures, and the practical application of statistical and machine learning techniques to commercial decision-making.

Lay Abstract

This report describes the work completed during a nine-month internship at ElectroRoute, a company that trades electricity generated by wind farms and solar farms on behalf of its clients. The company operates in the Irish and British electricity markets, where electricity is bought and sold through a series of auctions held at different times before it is actually needed.

The first project involved building an interactive online dashboard that allows the trading team to monitor how well they are performing. The dashboard displays charts and statistics covering areas such as income, energy production, and trading costs, and was designed so that new features can be added easily after the internship ends.

The second project analysed how the company trades its solar energy. Currently, most solar electricity is sold in the earliest auction, but the analysis investigated whether selling some of it in later auctions — when forecasts of how much electricity the solar farms will produce are more accurate — could lead to better financial outcomes. The analysis found that forecast errors and grid curtailment were the biggest factors affecting performance, and proposed a data-driven approach to deciding how to split volume across auctions each day.

The third project developed a regression model that combines two existing electricity demand forecasts to produce a single, more accurate prediction. By learning from historical data which forecast tends to be more accurate under different conditions, the model was able to reduce prediction errors compared to using either forecast on its own.

Alongside these technical projects, a practical trading strategy was developed and tested through simulated trading. Each day, weather forecasts, energy supply and demand data, and market conditions were reviewed to decide how much electricity to sell and when. The simulated trades were then compared against actual market outcomes, providing immediate feedback that was used to refine the strategy over time. This exercise built practical experience in making trading decisions under uncertainty — a skill that is difficult to develop through analysis alone.

Together, these projects and activities contributed practical tools, analysis, and market knowledge to support the company's trading operations, while providing hands-on experience in data analysis, programming, energy markets, and machine learning.

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It should be noted that a number of details have been redacted from this report in order to protect ElectroRoute's intellectual property and confidential client data. This includes forecast provider names, asset site names, trader names, generation figures, and commercially sensitive revenue and balancing cost data.

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List of Abbreviations

API	Application Programming Interface
BM	Balancing Market
BSRM	Balancing Services Revenue Model
CEO	Chief Executive Officer
CPPA	Corporate Power Purchase Agreement
CSS	Cascading Style Sheets
DA	Day-Ahead
DAM	Day-Ahead Market
ECC	European Commodity Clearing
EFET	European Federation of Energy Traders
ESB	Electricity Supply Board
GB	Great Britain
GW	Gigawatt
HTML	HyperText Markup Language
I-SEM	Integrated Single Electricity Market
IDA	Intraday Auction
IDA1	Intraday Auction 1
IDA2	Intraday Auction 2
IDA3	Intraday Auction 3
IDC	Intraday Continuous
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MCV	Market Clearing Volume
MOUG	Market Operator User Group
MW	Megawatt
MWh	Megawatt hour
NIV	Net Imbalance Volume
NSI	Normalised Solar Index
PnL	Profit and Loss
PPA	Power Purchase Agreement
REFIT	Renewable Energy Feed-in Tariff
REMIT	Regulation on Energy Market Integrity and Transparency
REST	Representational State Transfer
RMSE	Root Mean Square Error
SCADA	Supervisory Control and Data Acquisition
SEM	Single Electricity Market
SEMO	Single Electricity Market Operator
SNSP	System Non-Synchronous Penetration
SOL	Solar
SONI	System Operator for Northern Ireland
SQL	Structured Query Language
TSO	Transmission System Operator
VPP	Virtual Power Plant

1 Introduction

1.1 Company Background

ElectroRoute Holdings Limited is an international energy trading and services company owned by the Mitsubishi Corporation. The company's mission is "to solve the commercial mechanics of a decarbonised energy system." By combining deep expertise in energy markets with cutting-edge digital infrastructure, ElectroRoute helps clients maximise the value of their renewable assets, navigate market volatility, and contribute to system-wide decarbonisation.

The company operates both proprietary trading and physical shift-based trading functions. Proprietary trading involves trading the firm's own capital to take positions in energy markets based on price signals, forecasts, and market fundamentals, with the objective of generating profit. Shift trading primarily relates to the physical trading and real-time management of electricity and gas positions. This involves the continuous, 24/7/365, execution of trades to balance physical supply and demand, manage forecast deviations, and respond to real-time system conditions. Shift traders are responsible for ensuring that physical delivery obligations are met, while managing imbalance risk and maintaining compliance with market rules.

As at March 2025, ElectroRoute had a turnover of €753.7 million and 124 employees. The company has offices in Dublin (headquarters), Letterkenny, London, and Tokyo. For the 2024/25 financial year, ElectroRoute managed over 2 GW of client assets and executed on average 5,200 trades per day. ElectroRoute currently holds a 3.5 GW portfolio of physical assets, and it is set to grow further, with 79 MW of renewables currently operating in the GB market expanding to 531 MW by FY26/27.

ElectroRoute offers a comprehensive range of energy trading and market services, including power, green certificate, and gas trading across wholesale markets; the structuring of Corporate Power Purchase Agreements (CPPAs) between renewable generators and corporate off-takers; fixed-price revenue guarantee structures that provide generators with revenue certainty; renewable balancing and market access services encompassing forecasting, nomination, and imbalance management; and battery and flexible asset optimisation across energy, balancing, and ancillary service markets.

1.2 Internship Role and Project Area

The internship was based within ElectroRoute's Trading Operations team, which sits at the core of the company's commercial activity. This team is responsible for the real-time and forward management of physical energy positions across electricity and gas markets, as well as the analysis and optimisation of renewable and flexible assets. The role operated at the intersection of energy systems, financial markets, and data-driven decision-making.

A key focus of the project area was physical trading and asset optimisation. This involved forecasting renewable generation and demand, monitoring system conditions, managing imbalances, and executing trades to ensure that physical delivery obligations were met in real time. Due to the variable nature of renewable generation, effective forecasting, monitoring, and rapid response to market conditions are critical to minimising imbalance costs and maximising asset value.

The internship also sat heavily within the domain of data analysis and automation. ElectroRoute maintains large databases populated with live and historical market, weather, and asset data. These datasets are used to produce dashboards, Key Performance Indicators (KPIs), trading lookaheads, and analytical tools that support both shift traders and proprietary trading teams. As a result, strong technical skills in Python, SQL, and Excel were essential within the project area.

Day-to-day responsibilities included performing daily market analysis on Irish and GB power markets, confirming SCADA availability and following up on outages, compiling and presenting weekly KPIs, preparing

trading lookaheads, reviewing constraint and outage reports, and attending SEMO market calls. Beyond these operational duties, the internship involved the completion of the pre go-live components of the ElectroRoute Academy trader training programme, covering electricity market structures, power systems, weather and forecasting, risk and compliance, and the operational responsibilities of the trading desk.

1.3 The Irish Integrated Single Electricity Market

The Integrated Single Electricity Market (I-SEM) is the wholesale electricity market for the island of Ireland, operated jointly by EirGrid (the Republic of Ireland’s Transmission System Operator, or TSO) and SONI (the System Operator for Northern Ireland) through the Single Electricity Market Operator (SEMO). The I-SEM replaced the previous SEM arrangements in October 2018, introducing a series of sequential market timeframes designed to align with European electricity market coupling frameworks.

The market operates through a sequence of trading stages, each progressively closer to real-time delivery:

- **Day-Ahead Market (DAM):** The primary forward market, operated as a local auction through SEMOpx. Market participants submit buy and sell orders for each trading period of the following day, and prices are determined through market coupling. The DAM is the most liquid auction stage and typically where the majority of volume is traded.
- **Intraday Auction 1 (IDA1):** A coupled SEM–GB auction. The order book opens at 23:00 on D-19 and closes at 17:30 on D-1, covering trading periods from 23:00 to 23:00 (a full 24-hour delivery window). Each trading period is 30 minutes in duration.
- **Intraday Auction 2 (IDA2):** A coupled SEM–GB auction. The order book opens at 23:00 on D-19 and closes at 08:00 on the delivery day, covering trading periods from 11:00 to 23:00 (a 12-hour delivery window).
- **Intraday Auction 3 (IDA3):** A local auction. The order book opens at 23:00 on D-19 and closes at 14:00 on the delivery day, covering trading periods from 17:00 to 23:00 (a 6-hour delivery window).
- **Intraday Continuous (IDC):** A continuous trading platform operated through the European ECC clearing framework, allowing bilateral trades to be executed on a pay-as-bid basis up to one hour before delivery.
- **Balancing Market (BM):** The real-time balancing mechanism operated by the TSO to resolve any residual imbalance between supply and demand. Any difference between a participant’s final traded position and their metered output is settled at the imbalance price, which can be significantly more volatile than day-ahead or intraday prices.

Auction	Order Book Opening	Order Book Closure	Trading Periods	Duration
IDA1	23:00 (D-19)	17:30 (D-1)	23:00 – 23:00	24 hours
IDA2	23:00 (D-19)	08:00 (D)	11:00 – 23:00	12 hours
IDA3	23:00 (D-19)	14:00 (D)	17:00 – 23:00	6 hours

Table 2: Intraday auction timelines in the I-SEM. All times are in Irish local time. Unlike the DAM, the trading period duration for each intraday auction is 30 minutes.

For renewable generators, the sequential structure of the I-SEM creates both opportunities and risks. Wind and solar output is inherently uncertain, and forecast accuracy typically improves as delivery approaches. Traders must therefore decide how to allocate volume across auction stages, balancing the liquidity and

price certainty of the DAM against the improved forecast accuracy available at later intraday auctions. Any residual mismatch between the final traded position and actual metered generation is settled in the balancing market, where the imbalance price can result in either a cost or a gain depending on the direction of the system imbalance.

The System Net Imbalance Volume (NIV) represents the net volume the TSO must procure through the balancing mechanism to balance the grid. Its direction determines whether a generator's residual imbalance is rewarded or penalised at settlement. Understanding and managing this imbalance exposure is central to the work of ElectroRoute's trading operations team and underpins much of the analysis presented in this report.

1.4 Report Structure

Section 2 describes the development of a reusable SQL query interface. Section 3 covers the design of a KPI dashboard for tracking trading desk performance. Section 4 presents analysis of solar trading strategy, including forecast accuracy and auction split backtesting. Section 5 details the development of a machine learning demand forecasting ensemble. Section 6 describes the trading strategy framework developed through a paper trading exercise. Section 7 outlines professional certifications and development activities completed during the internship.

2 SQL Query Development

A task completed early in the internship was the development of a reusable SQL query function to efficiently retrieve data from ElectroRoute's market intelligence database. The database follows a star schema architecture, with a central fact table storing timestamped values and a series of dimension tables describing the category, country, source, zone, segment, and time interval of each record. Querying this database directly requires joining across multiple dimension tables, resulting in verbose and repetitive SQL statements that are prone to error when written manually for each new analysis.

To address this, a Python wrapper function was developed that abstracts the underlying query complexity into a simple, structured interface. The function accepts a set of descriptive parameters — category, country, source, zone, segment, date range, and an optional time interval — and dynamically constructs and executes the appropriate SQL query, returning the results as a time-indexed Pandas DataFrame ready for analysis.

```
def sql_query(category, country, source, zone, segment, start, end,
              interval=None):
    """
    Query market intelligence database for energy market and production data.

    Parameters:
    -----
    category : str
        Data category (e.g., "Market/DayAhead/Price",
        ↪ "Production/Renewable/Wind")
    country : str
        Two-letter country code (e.g., "IE", "GB")
    source : str
        Data source (e.g., market operator, forecast provider)
    zone : str
        Geographic or market zone
    segment : str
        Data segment (e.g., "Power")
    start : datetime or str
```

```

    Start datetime
end : datetime or str
    End datetime
interval : str, optional
    Time interval (e.g., "30 Minute", "Hourly"). If None, returns all
    ↪ intervals.

Returns:
-----
pd.DataFrame
    Time-indexed DataFrame with quantity values
"""

```

The function dynamically constructs the SQL query based on whether an interval parameter is provided. When specified, an additional join and filter is applied to the time interval dimension table; when omitted, all available intervals are returned. This flexibility allows the same function to be used for both half-hourly settlement data and hourly or daily aggregations without modification.

```

# Build query dynamically based on whether interval is provided
base_query = """
SELECT
    f.UTC_Datetime AS Start_Time,
    f.Value         AS quantity
FROM fact_table f
JOIN dimension d   ON f.dim_id   = d.id
JOIN dim_country c ON c.id       = d.country_id
                    AND c.code    = %s
JOIN dim_source s  ON s.id       = d.source_id
                    AND s.name     = %s
JOIN dim_zone z    ON z.id       = d.zone_id
                    AND z.name     = %s
JOIN dim_segment sg ON sg.id      = d.segment_id
                    AND sg.name    = %s
"""

if interval:
    base_query += """
JOIN dim_interval ti ON ti.id      = d.interval_id
                    AND ti.interval = %s
"""
    params = (category, country, source, zone, segment, interval, start,
              ↪ end)
else:
    base_query += """
JOIN dim_interval ti ON ti.id      = d.interval_id
"""
    params = (category, country, source, zone, segment, start, end)

base_query += """
WHERE d.category = %s
      AND f.UTC_Datetime >= %s
      AND f.UTC_Datetime < %s
ORDER BY f.UTC_Datetime
"""

```

Date parameters are automatically converted from Python `datetime` objects to string format if necessary, and the database connection is managed using externally stored credentials that are not exposed in the

codebase. The query results are returned as a Pandas DataFrame with the timestamp column converted to a DatetimeIndex, making the output immediately compatible with downstream analysis and plotting workflows.

```
# Convert datetime objects to strings if needed
if isinstance(start, (dt.date, dt.datetime)):
    start = start.strftime("%Y-%m-%d %H:%M:%S")
if isinstance(end, (dt.date, dt.datetime)):
    end = end.strftime("%Y-%m-%d %H:%M:%S")

# Connection parameters stored externally (not in code)
conn = connect_to_database()
cur = conn.cursor(dictionary=True)
cur.execute(base_query, params)
data = cur.fetchall()
cur.close()
conn.close()

df = pd.DataFrame(data)

if not df.empty and 'Start_Time' in df.columns:
    df['Start_Time'] = pd.to_datetime(df['Start_Time'])
    df = df.set_index('Start_Time')

return df
```

This function was used extensively throughout all three projects presented in this report, serving as the primary data retrieval interface for the KPI dashboard, solar trading analysis, and demand forecasting ensemble. By standardising how data was queried, it reduced the risk of errors in query construction, improved code readability, and significantly accelerated the development of new analyses. An example call to retrieve half-hourly day-ahead prices for Ireland is shown below:

```
da_prices = sql_query(
    category = "Market|DayAhead|Price",
    country = "IE",
    source = "MarketOperator",
    zone = "SEM",
    segment = "Power",
    start = "2025-04-01",
    end = "2026-04-01",
    interval = "30 Minute"
)
```

3 KPI Dashboard Build

At the start of the internship, I was tasked with building a KPI script to track the trading desk's performance. What began as a relatively straightforward scripting task gradually grew in scope, eventually evolving into a fully interactive web-based dashboard. The dashboard was built using Flask as the backend framework, with Plotly handling chart generation and a REST API architecture serving data to the frontend dynamically.

The final product provides a comprehensive view of portfolio performance across three asset groups — DAM, RFIT, and SOL — allowing users to filter by date range and aggregation frequency. Key metrics such as net income, capacity factor, dispatch down rates, and DA balancing costs are displayed through a series of interactive charts and summary widgets, giving the trading desk a clear view of operational performance.

A key milestone of the project was presenting the completed dashboard to the CEO and wider colleagues, which meant the final product needed to be both technically robust and visually accessible to a non-technical audience.

3.1 Dashboard Start-Up Process

The dashboard is built using Flask, a lightweight Python web framework, serving a single-page application with a REST API backend. Data is loaded on demand and persisted in a global state object for the session.

```
dashboard_state = {
    'results'      : None,
    'start_date'   : None,
    'end_date'     : None,
    'agg_freq'     : 'D',
    'last_updated' : None
}
```

The user selects a date range and aggregation frequency (Daily/Monthly) via the frontend, which triggers a POST request to `/api/load_data`. This loads all data into the global state, making it available to all subsequent API calls without re-querying the database.

The application exposes a series of REST API endpoints, each serving a specific function:

Route	Method	Purpose
<code>/api/load_data</code>	POST	Loads data for the selected date range
<code>/api/chart/<chart_id></code>	GET	Generates and returns a specific chart
<code>/api/stats/<group></code>	GET	Returns KPI summary statistics
<code>/api/status</code>	GET	Returns current dashboard state

Table 3: Flask REST API endpoints.

Charts are generated server-side using Plotly, serialised to JSON and passed to the frontend for rendering. The chart route handles both single and multiple figure responses dynamically, as some dashboard sections return more than one chart simultaneously.

Project Structure:

```
1 kpi_dashboard/
2 |─ app.py           # Flask app (main entry point)
3 |─ data_loader.py   # Data loading functions
4 |─ templates/
5 |   └─ index.html   # Dashboard UI
6 |─ static/
7 |   └─ *.png        # Logos
8 |─ charts/
9 |   └─ __init__.py   # Chart registry
10 |    └─ colors.py    # Color theme
11 |    └─ *.py         # Chart modules
```

Figure 1: Dashboard project directory structure.

The project was split into four main files and directories. `app.py` handles all Flask routing and API logic, `data_loader.py` contains all data loading and processing, and the `charts/` directory holds all

chart modules, with each individual .py file representing a self-contained chart. The frontend lives in the `templates/` and `static/` directories.

This structure was intentional — adding a new chart simply requires creating a new module in the `charts/` directory and registering it, without touching any of the core application logic. To ensure the dashboard could be maintained after the internship, a page was created on the Trading Operations Confluence covering how to run the dashboard, an overview of the project structure, and step-by-step instructions for adding new charts and widgets.

The screenshot shows a Confluence page titled "KPI Dashboard Documentation" by Finn O'Connor. The page includes a header with the ElectroRoute logo, a search bar, and a "+ Create" button. Below the header, the page title is "KPI Dashboard Documentation". The content starts with a section "Running the KPI Dashboard:" followed by a paragraph explaining that the KPI dashboard is a Flask-based dashboard for KPI analysis and portfolio performance. It then lists requirements and dependencies, followed by two code blocks: one for installing Flask and another for running the application. A terminal screenshot shows the command to run the application and the resulting output, including a warning about using a development server and the URL to access the dashboard.

By Finn O'Connor 3 min 12 Add a reaction No workflow

Running the KPI Dashboard:

The KPI dashboard is a Flask-based dashboard that allows for easy KPI analysis and overall portfolio performance. It is built in Python and HTML, and queries the database using SQL.

In order to run the dashboard, there are some requirements and dependencies that must be fulfilled on your computer. Ensure that Flask is installed.

```
1 Python 3.10+
```

```
1 pip install Flask==3.1.3
```

To run the dashboard, open a blank PowerShell terminal and paste the following to access the file directory:

```
1 cd "C:\usr\Code\trading_ops\python_3_consolidated\kpi\KPI Dashboard\kpi_dashboard"
```

To run the application, run the 'app.py' script using the following:

```
1 python app.py
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

Loading personal and system profiles took 4394ms.
(py3.10) C:\Users\oconnor> cd "C:\usr\Code\trading_ops\python_3_consolidated\kpi\KPI Dashboard\kpi_dashboard"
(py3.10) C:\usr\Code\trading_ops\python_3_consolidated\kpi\KPI Dashboard\kpi_dashboard [python_3.10 ~ +0 ~4 ~0 !]> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 109-280-487
```

Figure 2: Confluence instructions page.

Step 1: Create the New Chart File [🔗](#)

Create `charts/new_chart_name.py`:

```
1  """My Chart description"""
2
3  import plotly.graph_objects as go
4  import pandas as pd
5  from .colors import ER_DARK_GREEN, ER_LIGHT_GREEN, ER_ORANGE, STANDARD_LAYOUT
6
7  # Required metadata
8  CHART_ID = "my_chart"
9  CHART_TITLE = "My Chart Title"
10 CHART_SECTION = "position_analysis" # See sections below
11 CHART_GROUPS = ["DAM", "RFIT", "SOL", "ALL"]
12 CHART_DESCRIPTION = "What this chart shows"
13
14
15 def create_empty_figure(message):
16     fig = go.Figure()
17     fig.add_annotation(text=message, xref="paper", yref="paper", x=0.5, y=0.5, showarrow=False)
18     fig.update_layout(template="plotly_white", height=400)
19     return fig
20
21
22 def create_chart(results, group, agg_freq, start_str, end_str):
23     """
24     Parameters:
25         results: dict with keys 'DAM', 'RFIT', 'SOL', each containing {'df': DataFrame}
26         group: 'DAM', 'RFIT', 'SOL', 'ALL', or 'MARKET'
27         agg_freq: 'D' (daily) or 'MS' (monthly)
28         start_str, end_str: date strings for title
29
30     Returns:
31         plotly Figure (or tuple of Figure + HTML string, or list of Figures)
32     """
33
34     # Get data
35     if group == "ALL":
36         # Combine groups
37         dfs = [results[g]["df"] for g in ["DAM", "RFIT", "SOL"] if results.get(g)]
38         if not dfs:
39             return create_empty_figure("No data")
40         df = pd.concat(dfs).groupby(level=0).sum()
41     else:
42         data = results.get(group)
43         if data is None:
44             return create_empty_figure(f"No data for {group}")
45         df = data["df"]
46
47     # Check columns exist
48     if "QMLF" not in df.columns:
49         return create_empty_figure("Missing column: QMLF")
50
51     # Create figure
52     fig = go.Figure()
53     fig.add_trace(go.Scatter(x=df.index, y=df["QMLF"], name="Production", line=dict(color=ER_DARK_GREEN)))
54
55     fig.update_layout(
56         title=f"<b>{group}</b> - My Chart ({start_str} to {end_str})</b>",
57         template=STANDARD_LAYOUT["template"],
58         hovermode=STANDARD_LAYOUT["hovermode"],
59         legend=STANDARD_LAYOUT["legend"],
60         height=500,
61         margin=dict(t=80, b=120, l=60, r=60),
62     )
63
64     return fig
```

Figure 3: Template chart module showing required metadata and `create_chart()` structure.

Once the dashboard is running on the local host address, the user is presented with the interface shown in Figure 4. From here, the analysis start and end dates can be selected using the date widgets, and the aggregation frequency can be toggled between Daily, Weekly, and Monthly. Once a date range is selected, clicking **Load Data** queries the database and loads all data into the global state, with a loading indicator displayed while the data is being fetched. Once loaded, clicking **Refresh Charts** re-renders all charts for the currently active tab without re-querying the database.

The dashboard is organised into six tabs — DAM, RFIT, SOL, All Groups, Trader Performance, and Market Analytics — each containing a set of charts and KPI widgets relevant to that asset group or analysis type. Daily aggregation gives the most granular view of performance, while Monthly aggregation smoothens out short-term noise and can be more useful for identifying longer-term trends.

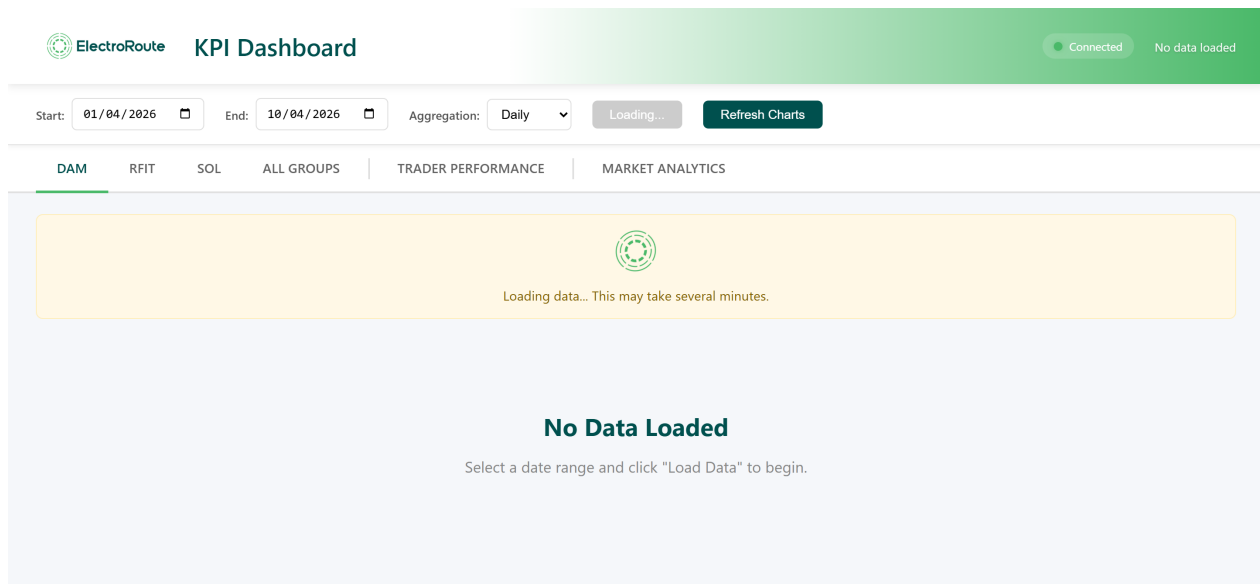


Figure 4: KPI dashboard interface with date range selector and aggregation toggle, loading DAM data.

3.2 Portfolio Performance

For the purposes of this report, the All Groups tab will be used to illustrate the dashboard’s functionality, rather than walking through each individual asset group tab. All Groups aggregates the results across DAM, RFIT, and SOL, giving a view of the entire portfolio’s performance in one place. The Portfolio Performance Summary widget displays the key metrics for the selected date range, including DA balancing cost, production, net income, capacity factor, dispatch down rate, total balancing cost, and curtailment. Below this, the Group Breakdown table splits each of these metrics by asset group, allowing the user to quickly identify which group is driving overall portfolio performance.

Capacity Factor:

```
capacity_factor = (metered_generation / installed_capacity) * 100
```

Dispatch Down Rate:

```
total_dispatch_down = curtailment + constraints
dispatch_down_rate = (total_dispatch_down / available_generation) * 100
```

Curtailment Rate:

```
curtailment_rate = (curtailed_volume / available_generation) * 100
```

Day-Ahead Balancing Cost per MWh:

```
da_balancing_cost = (metered_generation * da_price) - net_income
da_balancing_cost_per_mwh = da_balancing_cost / metered_generation
```

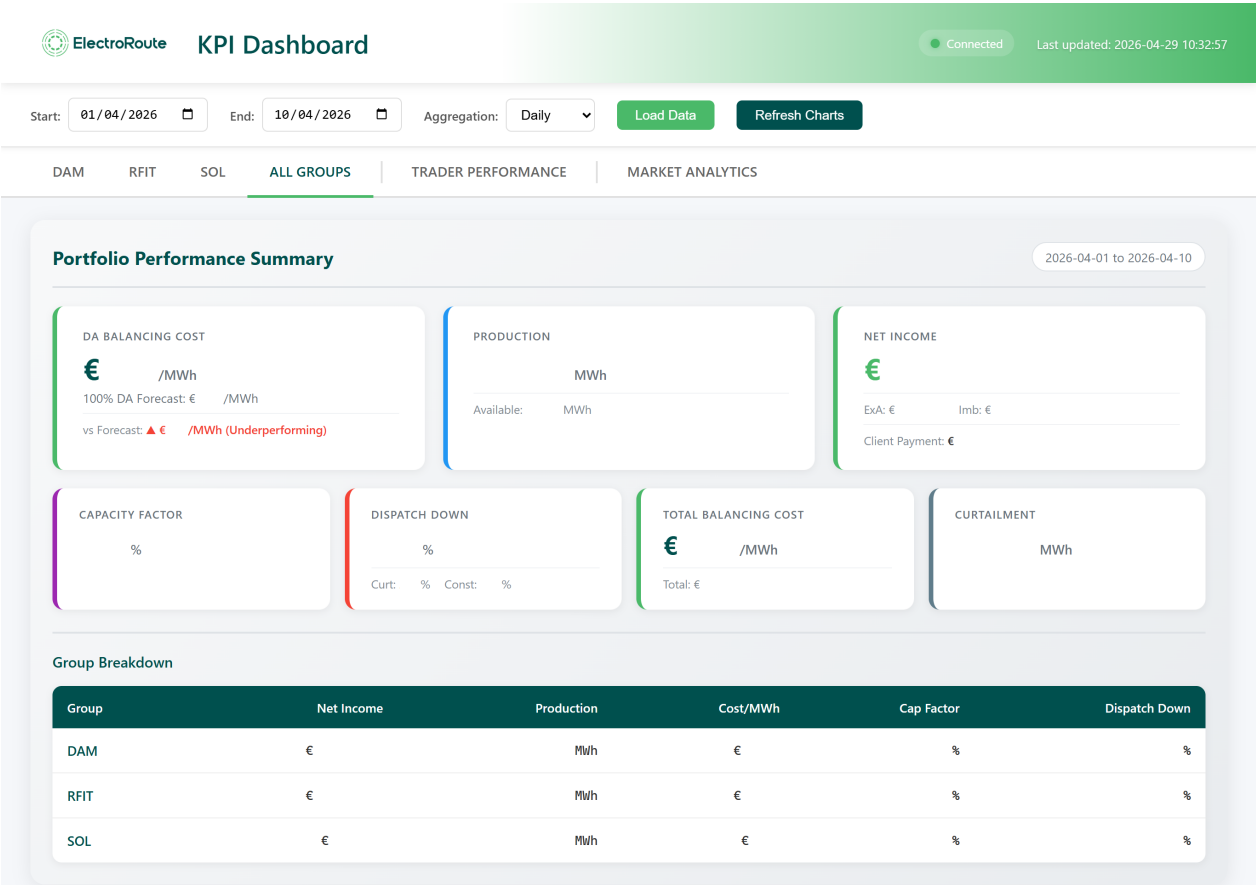


Figure 5: Portfolio performance summary and group breakdown table.

Portfolio Daily Balancing Costs: This chart displays the daily balancing costs across the selected date range, grouped by scenario for each day. Each cluster of bars represents a single day, with each bar corresponding to a scenario at 100%, 90%, and 110% of the base DA forecast, alongside the DA Position Cost and the Actual Balancing Cost. Costs can be both positive and negative, reflecting the nature of imbalance settlement, where a favourable imbalance position can result in a net gain rather than a cost. This view allows the user to quickly identify individual days where the actual cost deviated significantly from the forecast scenarios.

Portfolio Cumulative Balancing Costs: This chart shows the running total of each scenario’s balancing cost over the selected date range. Each line represents a different scenario, allowing the user to track how the actual balancing cost is trending relative to the forecast scenarios over time. A cumulative actual line that sits below the forecast scenario lines indicates that the portfolio has incurred lower balancing costs than anticipated, while a line trending above suggests costs have exceeded expectations.

Balancing Cost Scenarios

Balancing Cost Scenarios

Compares actual balancing costs against forecast scenarios (100%, 90%, 110% DA)

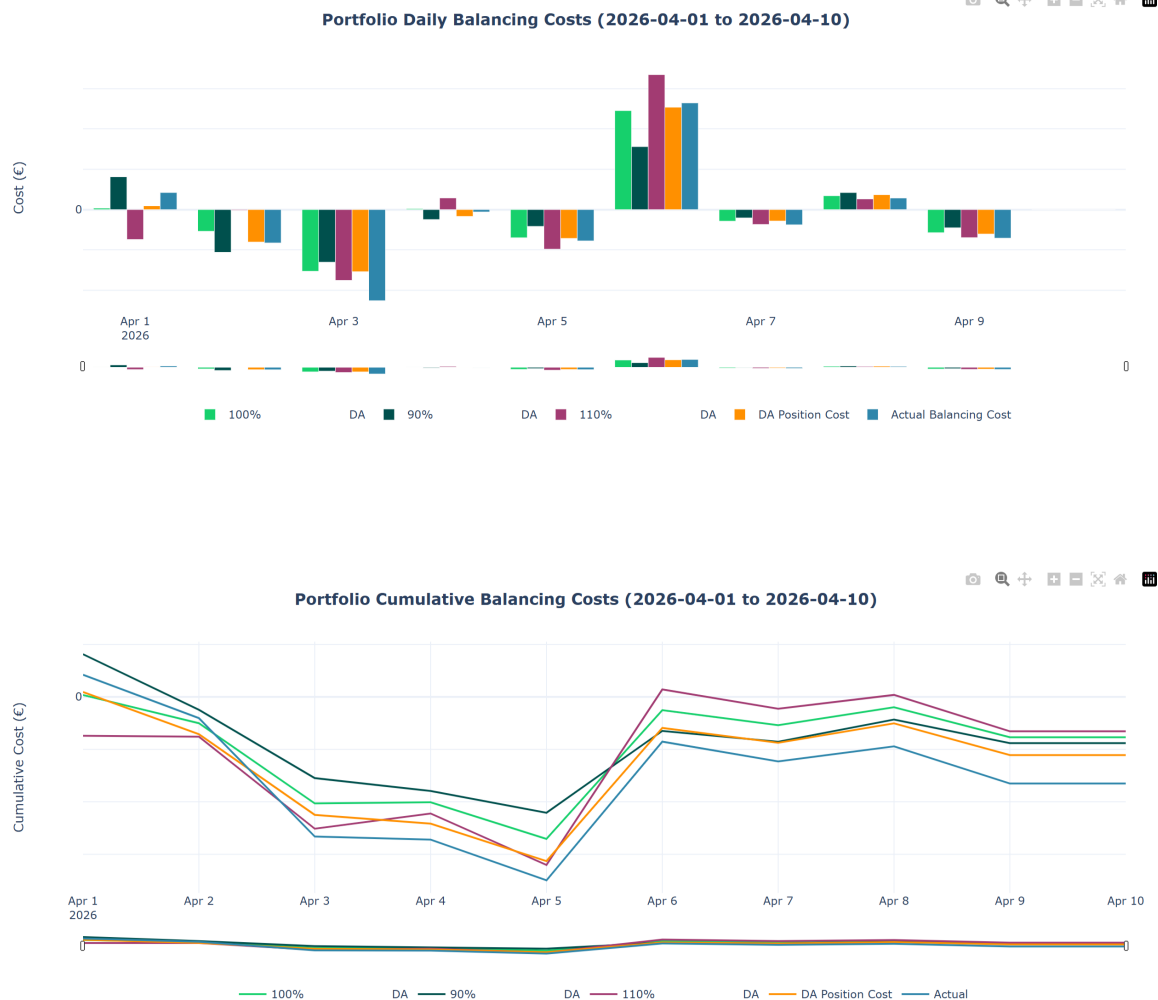


Figure 6: Daily and cumulative portfolio balancing costs by forecast scenario.

Portfolio Performance vs DA Forecast: This chart combines a bar and line series to illustrate how the actual balancing cost compares to the 100% DA forecast scenario on both a periodic and cumulative basis. The bars represent the difference between the actual balancing cost and the forecast for each individual day, coloured orange where the actual cost exceeded the forecast and green where the actual cost came in below it. The cumulative difference line tracks the running total of these daily differences over the selected date range, giving the user a view of overall performance.

A cumulative line trending downward and into negative territory indicates that the portfolio is consistently outperforming the forecast, meaning actual balancing costs are lower than expected. A line trending upward indicates the opposite. The summary table beneath the chart provides a breakdown of the total cost for each scenario over the period, along with the difference relative to the actual balancing cost, allowing the user to assess how each forecast scenario compared to what was ultimately settled.

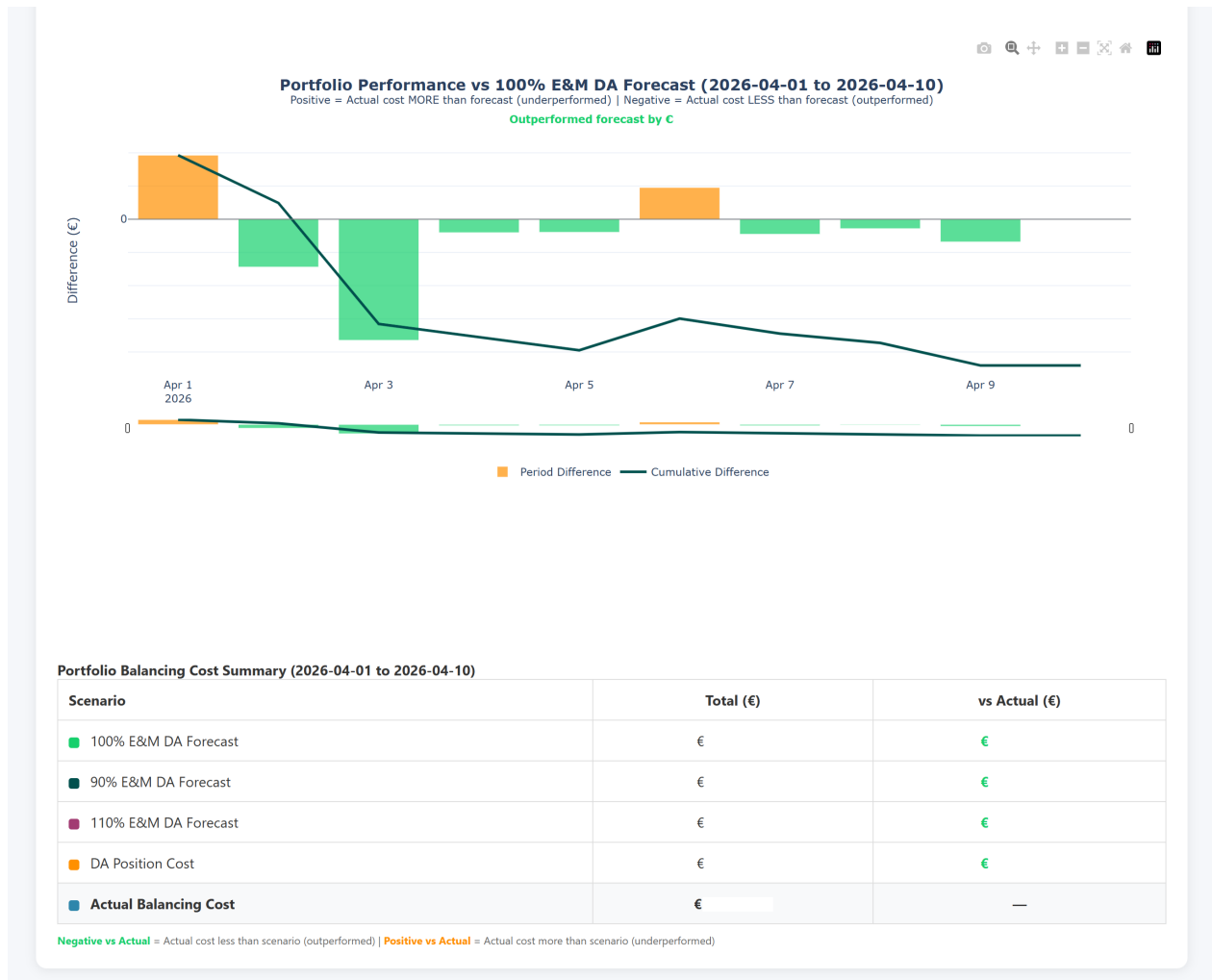


Figure 7: Actual vs. DA forecast balancing cost with cumulative difference.

Dispatch Down Analysis: The volume chart breaks down available generation into its constituent components across the selected date range. Metered generation, curtailment, and network constraints are stacked together, with the black line overlaid representing total available generation. The difference between the black line and the top of the green area represents the volume dispatched down, split between curtailment in orange and constraints in red.

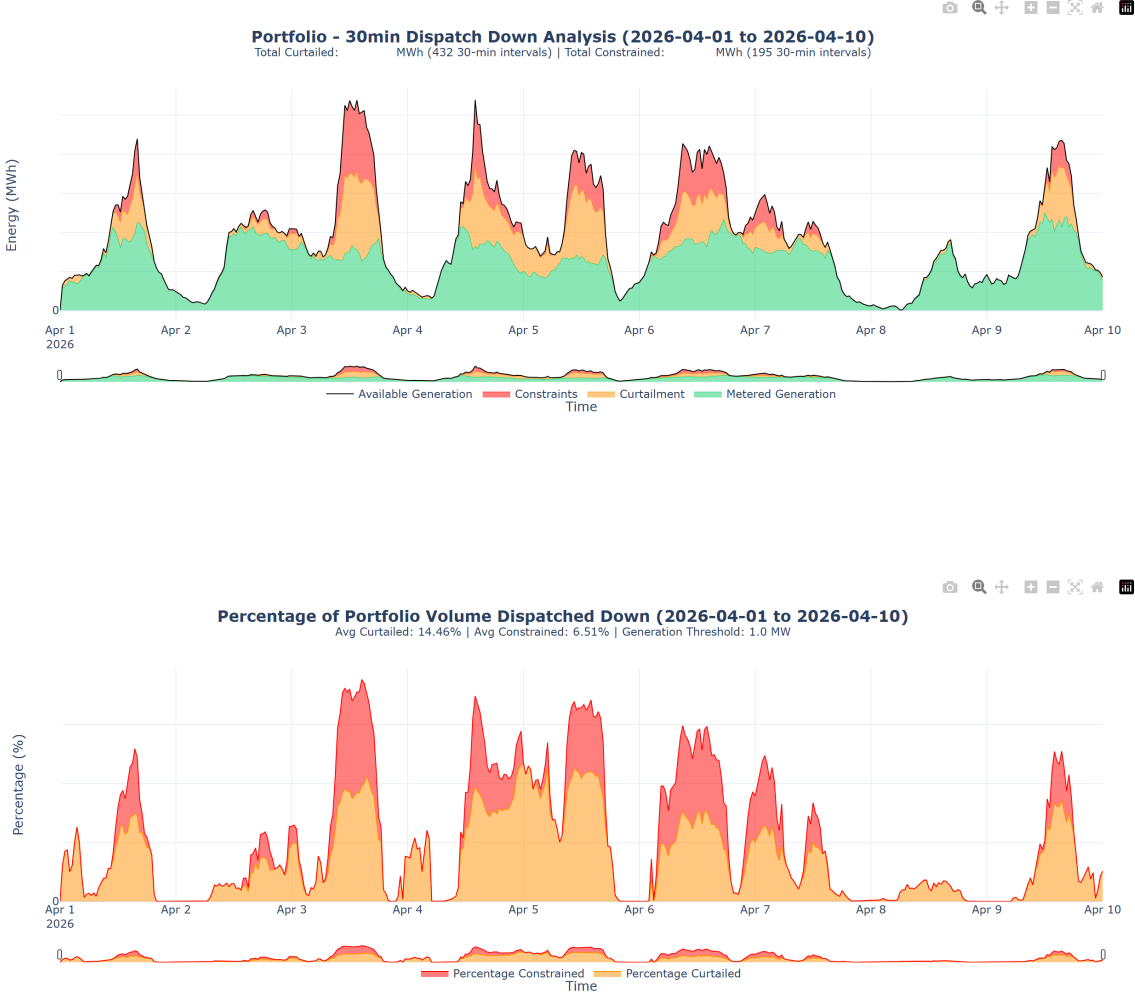
Percentage of Volume Dispatched Down: Rather than showing absolute volumes, this chart expresses curtailment and constraints as a percentage of available generation for each period, allowing the user to assess the relative severity of dispatch down events independently of the overall generation level. Percentage values are only calculated for periods where generation exceeds a minimum threshold, filtering out periods of meter noise or standby operation.

Dispatch Down Summary Table: The summary table provides a consolidated view of key dispatch down metrics over the selected date range, reporting the total volume lost to curtailment and constraints, the percentage of available generation dispatched down, the average percentage during periods of active generation, and the number of settlement periods affected.

Curtailment

Dispatch Down Analysis

Volume and percentage analysis of curtailment and constraints on generation



Portfolio Dispatch Down Summary (2026-04-01 to 2026-04-10)

Metric	Curtailment	Constraints	Total
Volume (MWh)			
% of Available Generation	21.82%	12.34%	34.16%
Avg % (when generating)	14.46%	6.51%	20.97%
Periods Affected	432 30-min intervals	195 30-min intervals	-
% of Periods Affected	99.8%	45.0%	-
Metered Generation	MWh		
Available Generation	MWh		
Periods with Generation (>1.0 MW)	432 / 433 (99.8%)		

Note: Percentage calculations exclude periods where generation is below 1.0 MW threshold (filters meter noise/standby).

Figure 8: Dispatch down volume breakdown, percentage analysis, and summary.

IDA Net Gain vs. Imbalance: This chart compares the cumulative gain or loss generated through IDA trading against what would have been achieved had those positions been left to settle at imbalance. The upper panel breaks this down by auction, showing the cumulative net gain for IDA1, IDA2, IDA3, and IDC individually over the selected date range. This allows the user to identify which auctions are contributing positively to trading performance and which are detracting from it. The lower panel aggregates all auctions into a single cumulative total, giving an at-a-glance view of whether IDA trading has been beneficial overall. A line trending upward and into positive territory indicates that trading across the intraday auctions has outperformed the imbalance settlement price, while a line trending negative indicates the opposite.

IDA Net Gain Summary Table: The summary table reports the total net gain attributed to each auction over the selected period, alongside each auction's contribution as a percentage of the overall total. This allows the user to quickly assess which auctions have driven the most value and whether any individual auction has been a consistent source of loss.

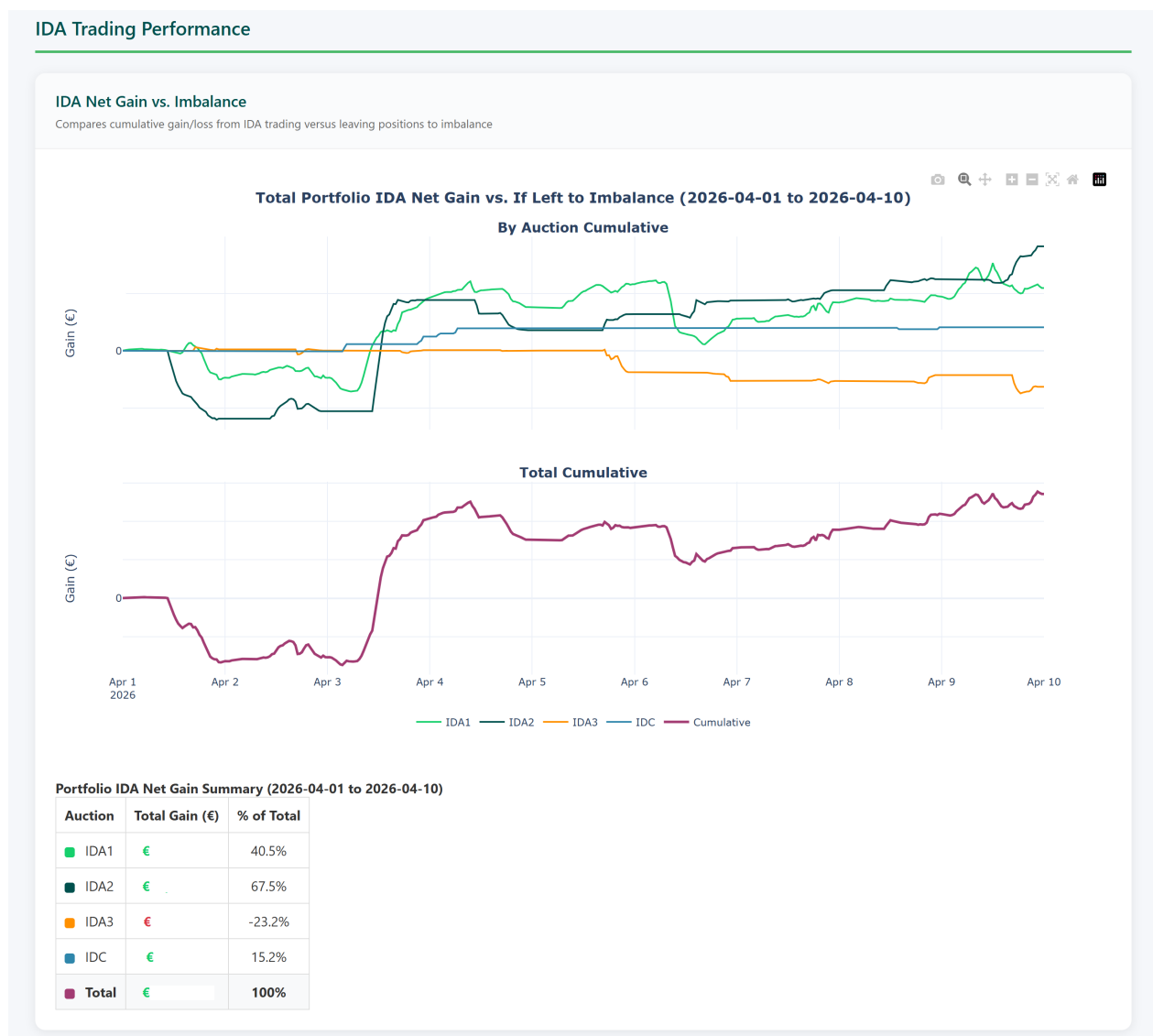


Figure 9: Cumulative IDA net gain vs. imbalance by auction and total.

Trade All Auctions: This chart compares cumulative balancing costs under two scenarios: actual trading decisions made across the period, and a simulation in which the full difference between each successive auction forecast and the current position is traded at every IDA auction, subject to available volume limits. The cumulative actual and simulated costs are plotted as lines, with the running difference shown as bars.

Positive bars indicate periods where trading all auctions would have reduced costs, while negative bars indicate that actual trading outperformed the simulation. The headline figure in the chart title summarises the total cost difference over the selected period, expressed in euros and as a percentage of actual balancing cost, for either the individual group or the combined portfolio depending on the view selected.

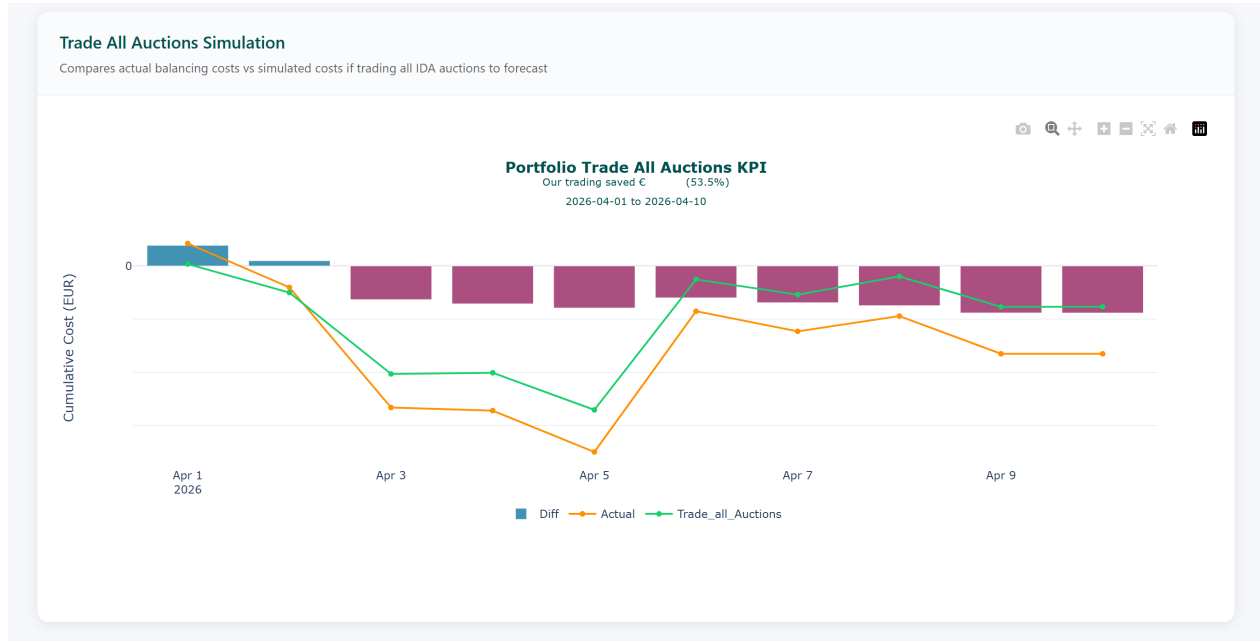


Figure 10: Actual vs. simulated “trade all auctions” cumulative balancing cost.

Summary Statistics: This section presents two exportable tables covering the selected date range. When viewing the portfolio, an upper table reports aggregate figures across all groups, and a lower table breaks the same metrics down by group (DAM, RFIT, SOL). When viewing a single group, a single table is shown for that group alone. All tables can be exported via the Plotly toolbar.

The metrics reported are defined as follows:

- **Ex-Ante Income (€):** Total income generated from day-ahead and intraday auction trades, calculated as traded volume multiplied by the corresponding auction price across all settlement periods.
- **Imbalance Income (€):** Income or cost arising from the residual imbalance position, being the difference between metered production and the final traded position, settled at the imbalance price. A negative value indicates a net imbalance cost.
- **Net Income (€):** The sum of ex-ante income and imbalance income, representing total revenue received over the period.
- **Client Payment (€):** Total payment made to the client over the period, calculated using the contracted reference price applied to metered production.
- **Production (MWh):** Total metered energy output across all assets in the group or portfolio over the selected period.
- **Capacity Factor (% , weighted avg):** Ratio of actual production to maximum possible production given installed capacity, weighted across assets by installed capacity.
- **Dispatch Down Rate (%):** Proportion of available generation that was not dispatched, decomposed into two sub-components:

- **Curtailment Rate (%)**: Share of lost generation attributable to grid operator curtailment instructions.
- **Constraint Rate (%)**: Share of lost generation attributable to network constraint actions.
- **DA Position Cost (€/MWh)**: Average price per MWh at which the day-ahead position was sold, reflecting the blended day-ahead auction price across the period. For RFIT assets this accounts for the applicable RFIT pricing mechanism rather than the standard day-ahead price.
- **100% DA Forecast Bal Cost (€/MWh)**: Hypothetical balancing cost per MWh that would have been incurred had the portfolio traded exactly its day-ahead forecast volume with no intraday adjustment.
- **90% DA Forecast Bal Cost (€/MWh)**: The same calculation applied to 90% of the day-ahead forecast volume, providing a downside sensitivity on forecast trading.
- **110% DA Forecast Bal Cost (€/MWh)**: The same calculation applied to 110% of the day-ahead forecast volume, providing an upside sensitivity on forecast trading.
- **Actual Balancing Cost (€/MWh)**: Realised balancing cost per MWh over the period, reflecting the actual imbalance position after all trading decisions across DA and IDA auctions.
- **Total Balancing Cost (€)**: Total absolute balancing cost over the period, equal to the actual balancing cost per MWh multiplied by total production.

The **Group Breakdown** table reports net income, production, cost per MWh, capacity factor, dispatch down rate, and total balancing cost separately for each group, allowing performance to be assessed at an individual portfolio level.

Summary Statistics (Exportable)

Detailed summary tables for data export. Use the Plotly toolbar to download as image or data.

Portfolio Summary (2026-04-01 to 2026-04-10)

Metric	Portfolio Total
Ex-Ante Income (€)	€
Imbalance Income (€)	€
Net Income (€)	€
Client Payment (€)	€
Production (MWh)	
Capacity Factor (% , weighted avg)	
Dispatch Down Rate (%)	
- Curtailment Rate (%)	
- Constraint Rate (%)	
DA Position Cost (€/MWh)	€
100% DA Forecast Bal Cost (€/MWh)	€
90% DA Forecast Bal Cost (€/MWh)	€
110% DA Forecast Bal Cost (€/MWh)	€
Actual Balancing Cost (€/MWh)	€
Total Balancing Cost (€)	€

Group Breakdown (2026-04-01 to 2026-04-10)

Group	Net Income	Production (MWh)	Cost (€/MWh)	Cap Factor (%)	Dispatch Down (%)	Total Cost
DAM	€		€			€
RFIT	€		€			€
SOL	€		€			€

Figure 11: Portfolio and group-level summary statistics.

Portfolio Positions: This chart displays the half-hourly net traded position in MWh for each group — DAM, RFIT, and SOL — over the selected date range. The net position is calculated as the sum of volumes traded across the day-ahead and all intraday auctions. The SOL series follows a characteristic solar generation profile, rising during daylight hours and returning to zero overnight, while the DAM and RFIT series reflect wind-dominated portfolios with less structured intraday variation. The chart allows the user to compare the scale and shape of each group’s traded position over time and identify periods of elevated or reduced market activity.

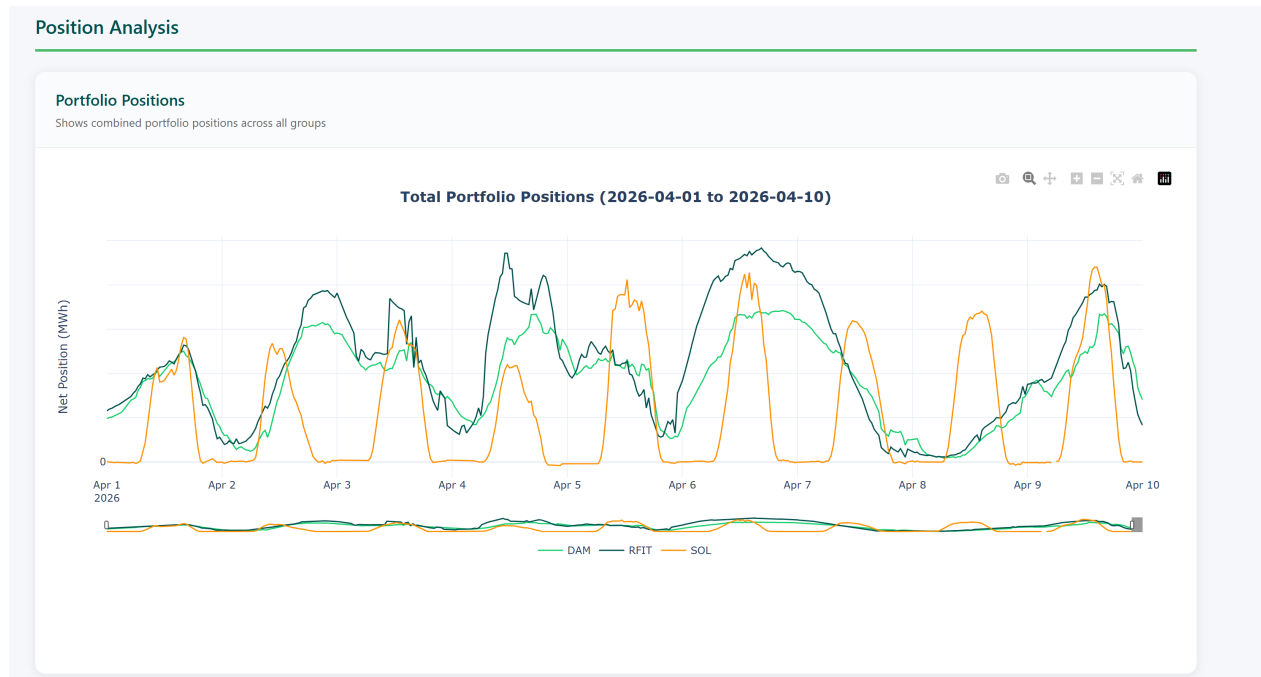


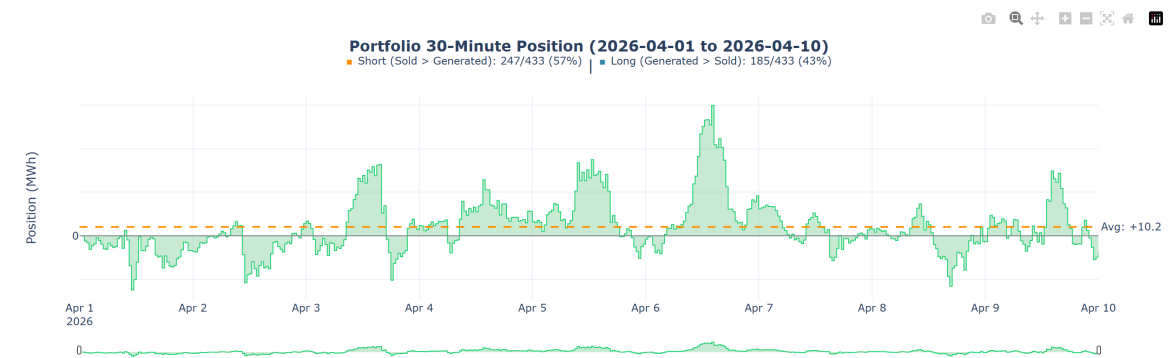
Figure 12: Half-hourly DAM, RFIT, and SOL traded positions.

Mean Long/Short Position: This chart displays the half-hourly net position, defined as the difference between total traded volume and metered generation at each settlement period. A positive value indicates a short position, where more volume was sold than generated, and a negative value indicates a long position, where generation exceeded traded volume. The dashed average line summarises the overall directional bias across the period. The accompanying summary table reports the number and proportion of settlement periods spent in each position, and classifies the overall trading bias accordingly. This is useful for identifying whether the portfolio has a systematic tendency to over- or under-trade relative to actual generation, with direct implications for imbalance exposure and balancing costs.

Portfolio Position vs System NIV: The System NIV represents the net volume the system operator must procure through the balancing mechanism to balance the grid, and its direction determines whether a generator’s residual imbalance is rewarded or penalised at settlement. This chart overlays the portfolio net position against the system NIV, giving a visualisation of how frequently the portfolio was on the right or wrong side of the system imbalance over the selected period, along with the resulting net imbalance value.

Mean Long/Short Position

Shows average position direction over time. Positive = Short (sold > generated), Negative = Long (generated > sold). Identifies systematic trading biases.



Portfolio Position Summary (2026-04-01 to 2026-04-10)

Metric	Value	Percentage
■ Short Periods (QEX > QMLF)	247 periods	57.0%
■ Long Periods (QMLF > QEX)	185 periods	42.7%
Total Periods	433 periods	100%
■ Average Position	+ MWh	Short Bias

Trading Bias: Tendency to sell more than generated

Figure 13: Half-hourly portfolio long/short position analysis.

Portfolio Position vs System NIV

Compares portfolio net position against system imbalance

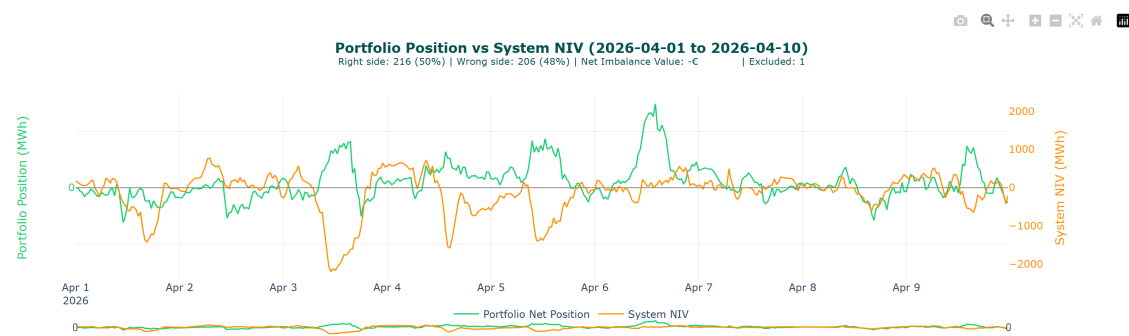


Figure 14: Portfolio net position overlaid against system NIV.

3.3 Trader Performance

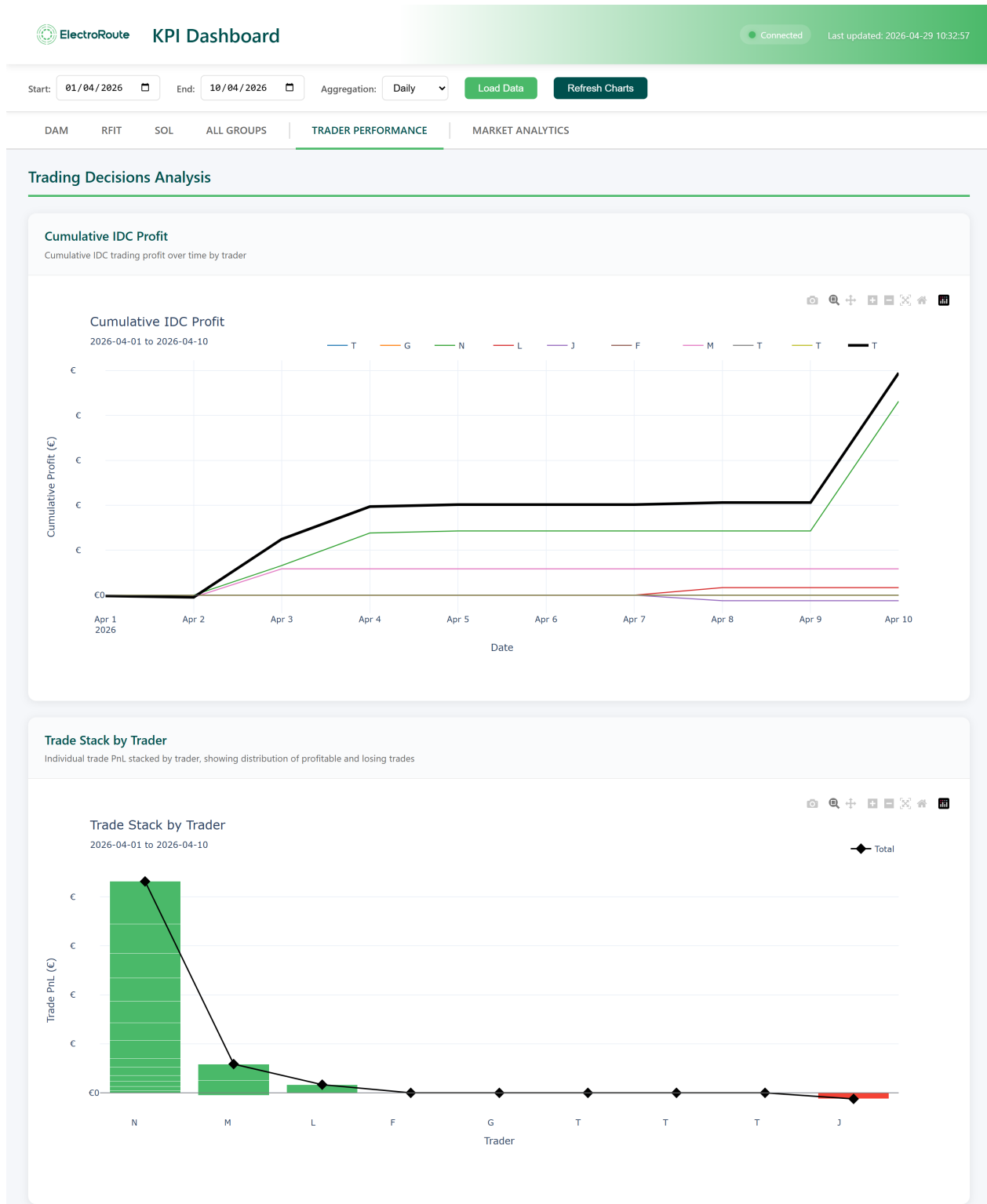
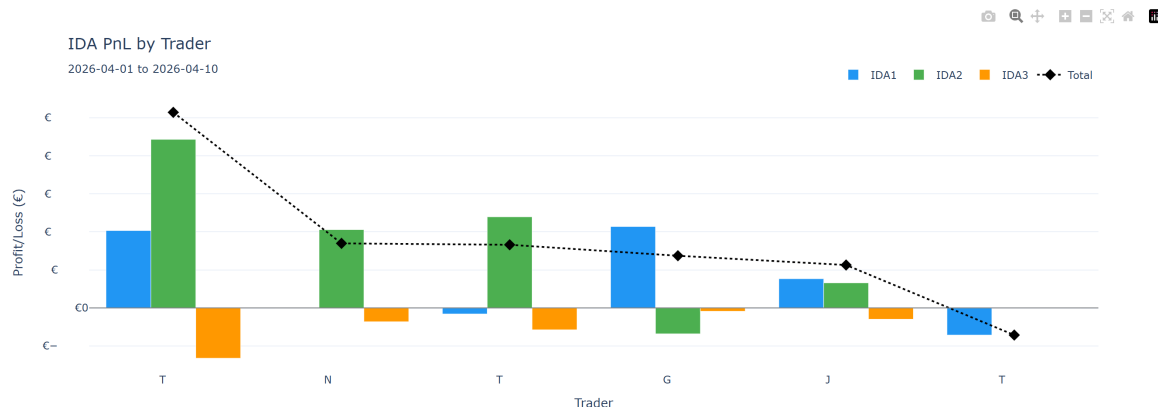


Figure 15: Cumulative IDC profit by trader and total trade stack PnL.

Trader Overview

IDA PnL by Trader

Breakdown of IDA1, IDA2, and IDA3 profit/loss by trader



IDA Trading Efficiency (€/MWh)

Trading efficiency measured as profit per MWh traded for each auction

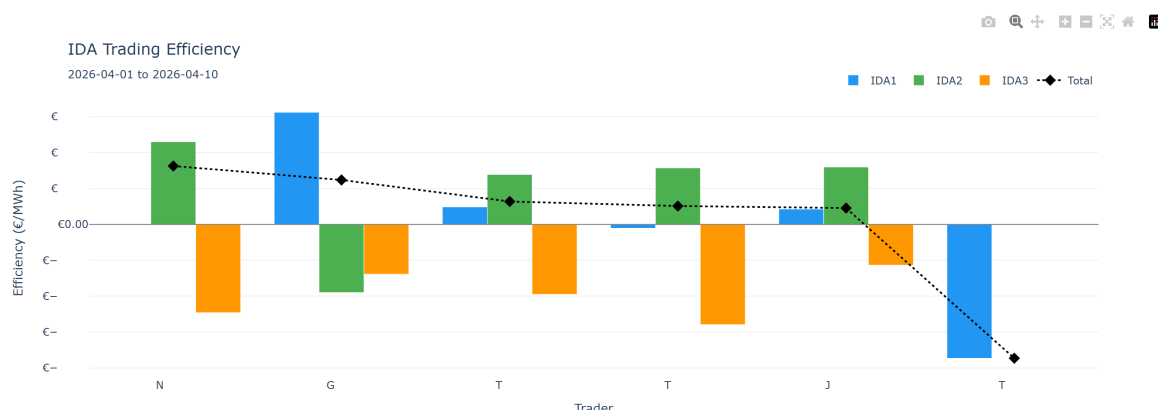


Figure 16: IDA PnL breakdown and trading efficiency (€/MWh) by trader.

Cumulative IDC Profit: This chart tracks each trader's intraday continuous trading profit cumulatively over time. It highlights whether returns are being generated consistently or concentrated in specific periods, and allows individual performance trajectories to be directly compared.

Trade Stack by Trader: This chart displays the total PnL contribution of each trader over the selected period. It highlights the distribution of positive and negative contributors, providing a clear ranking of individual impact on overall portfolio performance.

IDA PnL by Trader: This chart displays each trader's profit and loss broken down by intraday auction session (IDA1, IDA2, IDA3). It highlights where value is being generated or lost at both an individual and auction level.

IDA Trading Efficiency — €/MWh: This chart normalises each trader's auction PnL by volume traded, expressing performance in €/MWh. It highlights whether returns are driven by skilled positioning or simply by trading large volumes.



Figure 17: IDC profit by trader and daily IDC profit with cumulative line.

IDC Profit by Trader: This chart displays the total intraday continuous trading profit for each trader over the selected period. It highlights the relative contribution of each individual to overall team performance, making it straightforward to identify the primary profit drivers.

Daily IDC Profit: This chart shows the team's daily intraday continuous trading profit as a bar chart, overlaid with a cumulative profit line. It highlights both the consistency of daily performance and the overall profit trajectory across the period, including the team's win rate.

3.4 Market Analytics

Forecast Analysis: This chart overlays DA, IDA1, IDA2, and IDA3 forecasts against actual metered generation, alongside a forecast error plot for each auction stage. It highlights how accuracy evolves as delivery approaches and identifies periods of systematic bias. An equivalent plot was produced for Forecast Provider B, enabling direct comparison between the two providers.

Auction Prices: This chart displays DA, IDA1, IDA2, IDA3, and imbalance prices over time, accompanied by a summary statistics table. It highlights price spread and volatility across auction stages, and identifies periods of significant divergence between markets.

Imbalance Price Volatility: This chart displays the percentage spread between the imbalance and DA price over time, accompanied by a summary statistics table. It highlights periods of elevated volatility where imbalance exposure carries greater financial risk, with low-price periods excluded to avoid distortion.

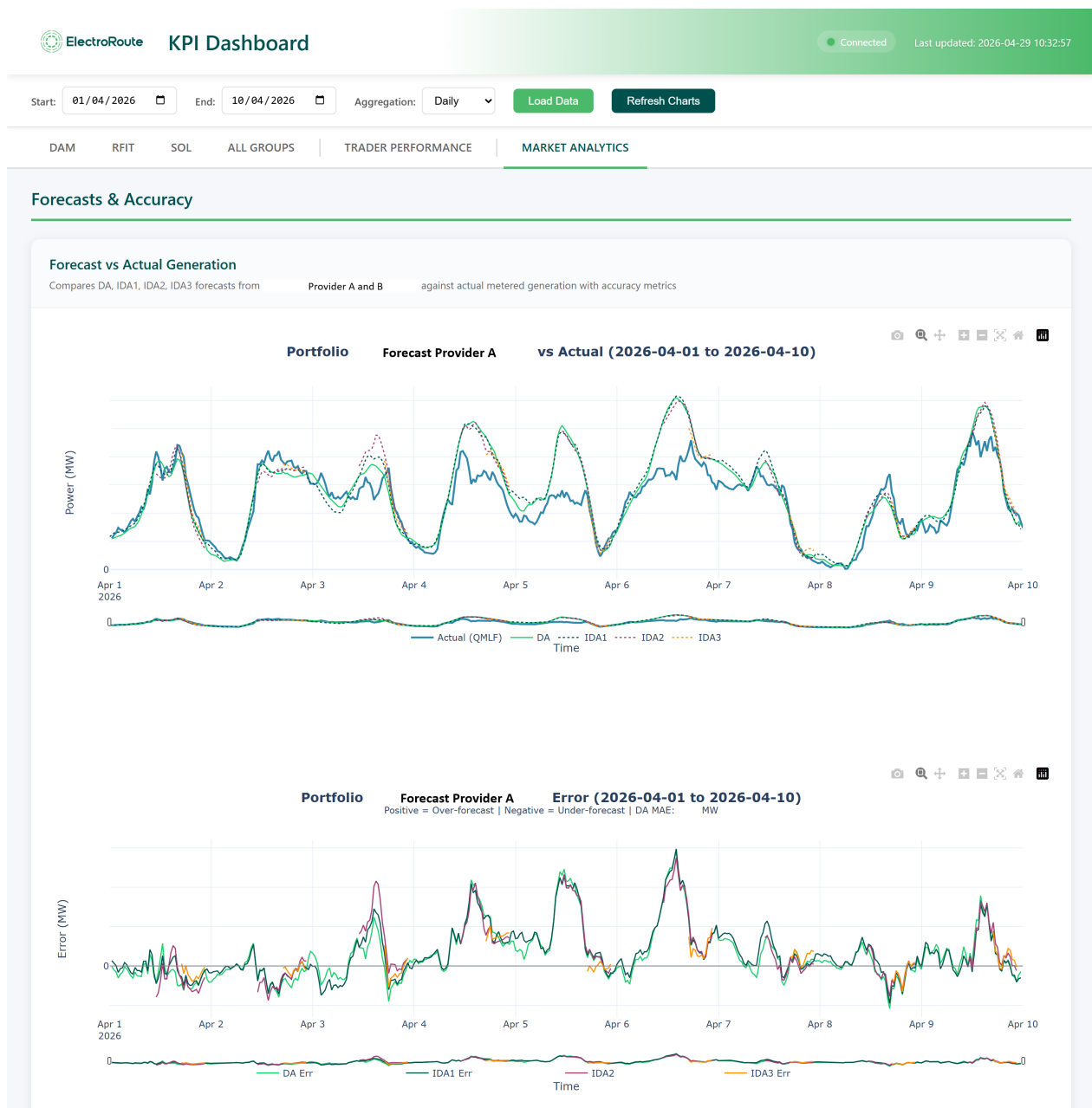
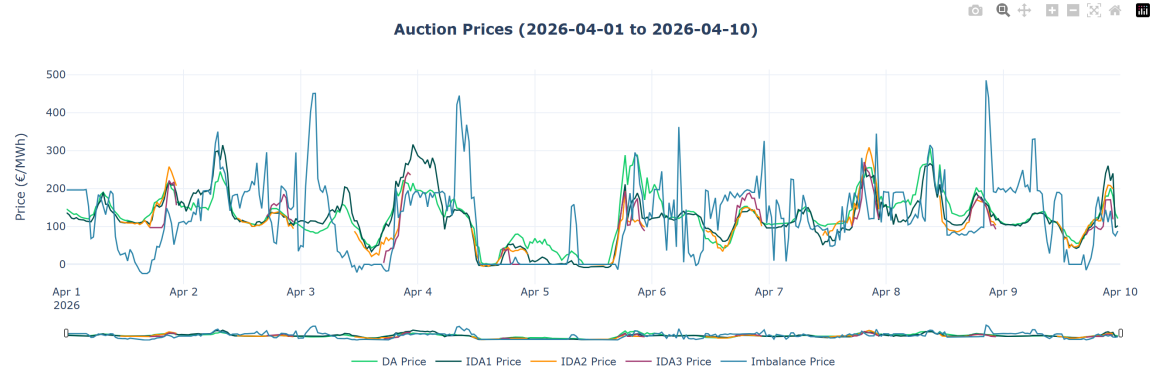


Figure 18: Forecast analysis comparing DA and IDA forecasts against metered generation.

Market Prices

Auction Prices

Shows DA, IDA1, IDA2, IDA3 and Imbalance prices over time

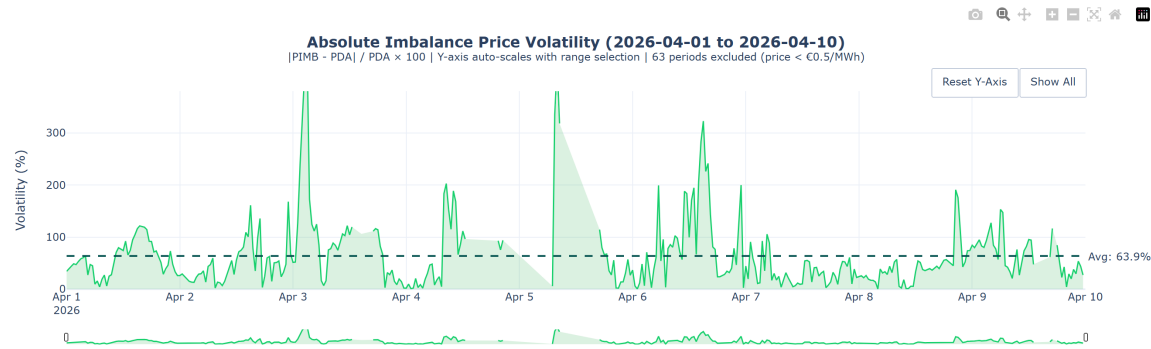


Price Statistics (2026-04-01 to 2026-04-10)

Price	Mean (€/MWh)	Min (€/MWh)	Max (€/MWh)	Std Dev
DA Price	€126.18	€0.00	€307.63	€59.23
IDA1 Price	€119.97	€-8.00	€316.00	€66.80
IDA2 Price	€102.14	€-5.00	€308.30	€62.86
IDA3 Price	€125.16	€0.00	€269.78	€63.02
Imbalance Price	€123.85	€-24.05	€485.36	€98.54

Imbalance Price Volatility

Shows imbalance price spread and volatility analysis



Imbalance Volatility Statistics (2026-04-01 to 2026-04-10)

Metric	Value
Average Volatility	63.86%
Maximum Volatility	447.09%
Minimum Volatility	0.37%
Standard Deviation	65.45%
Excluded Periods	63 of 433 (price < €0.5/MWh)

Figure 19: Auction price comparison and imbalance price volatility analysis.

3.5 Skills and Learning Outcomes

The KPI dashboard project developed a broad range of technical skills across full-stack web development, data engineering, and visualisation. Designing and building the Flask backend provided hands-on experience with REST API architecture, server-side routing, and managing application state, while constructing the frontend in HTML, CSS, and JavaScript developed an understanding of how data is consumed and rendered in a browser environment. Working with Plotly for chart generation strengthened my ability to produce clear, interactive visualisations that communicate complex data effectively to both technical and non-technical audiences.

From a data engineering perspective, the project required writing extensive SQL queries to extract and join data from multiple database tables, and building a robust data processing pipeline in Python using Pandas to clean, aggregate, and transform raw market data into the metrics and structures required by each chart. Designing the modular chart architecture — where each chart is a self-contained module that can be added or removed without modifying the core application — developed an appreciation for writing maintainable, scalable code that others can work with after handover.

Presenting the completed dashboard to the CEO and wider team was a significant professional milestone. Preparing for this presentation required careful consideration of how to frame technical work for a non-technical audience, focusing on the business value delivered rather than implementation detail. The experience reinforced the importance of clear communication and the ability to tailor the depth and language of a presentation to the audience.

Finally, producing comprehensive documentation on Confluence to support the dashboard's ongoing maintenance after the internship developed technical writing skills and reinforced the importance of knowledge transfer in a professional environment. Overall, this project provided the most sustained and technically diverse learning experience of the internship, spanning backend development, frontend design, database querying, data processing, visualisation, presentation, and documentation.

4 Solar Trading Strategy Development

This section presents analysis conducted in collaboration with a senior trader to explore more effective approaches to trading the company's solar assets. The current strategy of selling the majority of solar positions at the day-ahead auction may not always be optimal, and this work investigates whether dynamically allocating volume across DA, IDA1, IDA2, and IDA3 auctions based on forecast accuracy and price signals could improve revenue outcomes. Strategies were back-tested across the last 12 months to identify the key drivers that influence which auction stage yields the best result.

The longer term goal is to develop a predictive model that determines the optimal volume split across auctions, giving traders a data-driven edge when allocating volume. The analysis was conducted in Python and SQL, with findings presented to the wider team on Thursday 30th April.



Objective:

- The overall objective of this project is to investigate how we could trade our solar positions better across different auctions.
- Currently, we have a tendency to just sell most of our solar positions at the day-ahead auction and either modify positions at later auctions or just settle our positions in the balancing market.
- This analysis investigates if this is the best strategy, and if there are other more strategic ways to manage our positions in order to maximise our revenue.
- Analysis and strategies were back tested across the last 12 months to see if we could have done things differently – and give a framework for how we could possibly change how we do things going forward.
- The first section of the slides focuses on forecasting accuracy analyses, followed by strategy.

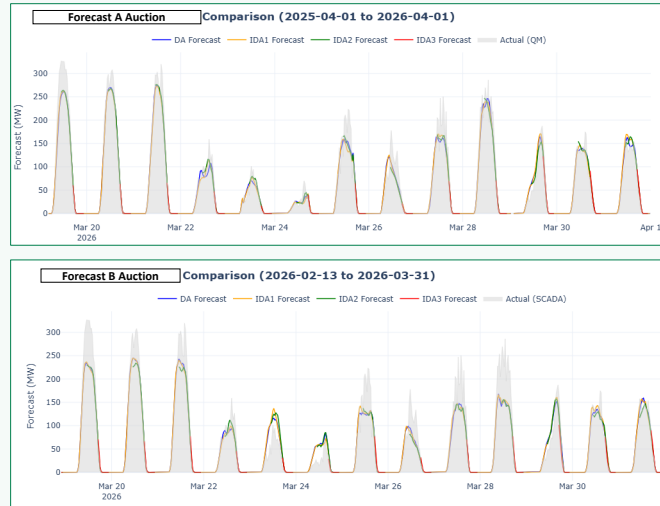


In order to once again protect any sensitive data, some of the details within the presentation were redacted. The first slide gave an introduction into what the following analysis would entail. Some of the key statistical measurements were defined for anyone who wasn't too familiar with them. The error was also defined to show the sign convention.

The total market clearing volumes for each auction were also compared with our traded activity to explore how much influence we have on the market.

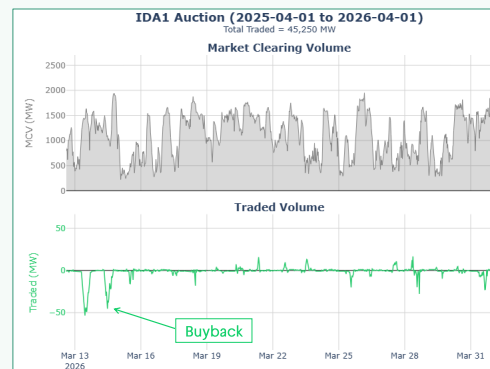
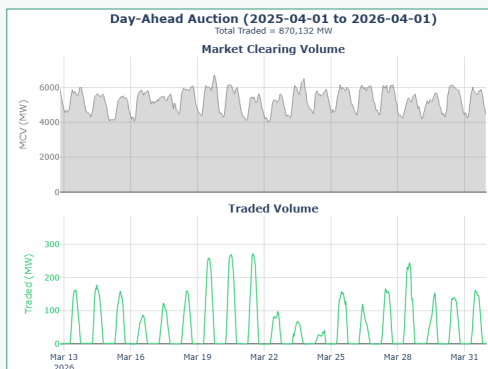
Solar Forecast Accuracy

- The following slides look at the performance and comparison of the two forecast providers, Forecast A vs Forecast B.
- Dispatch curtailment and constraint volumes were ignored as the generation forecasts don't take dispatch down into consideration.
- Some of the following key indicators are used to quantify forecast accuracy:
 - ME/MBE = mean error/bias
 - MAE = mean absolute error
 - RMSE = root mean squared error
 - MAPE = mean absolute percentage error
- Error Sign Convention:**
- Error = Forecast – Metered Generation**
 - Negative = Long/Under-forecasting
 - Positive = Short/Over-forecasting



Current Auction Trading Volumes

Auction	MCV (MW)	Traded (MW)	Sold (MW)	Bought (MW)	% of MCV	Buyback %
Day-Ahead	44,459,418.4	870,131.6	867,190.8	2,940.8	1.96	0.34
IDA1	20,077,865.5	45,250.5	14,503.0	30,747.5	0.23	67.95
IDA2	2,225,002.5	27,731.1	10,083.2	17,647.9	1.25	63.64
IDA3	515,106.0	5,762.9	1,995.8	3,767.1	1.12	65.37



4.1 Filtering the Data

Before conducting the analysis, the forecast data required filtering as it included many periods of zero generation such as night time periods. Initially a static time-based mask was applied, defining fixed windows for each auction stage during which solar generation was expected to occur. However, due to the solar path shifting throughout the year, the periods of active generation vary significantly across seasons, making a fixed mask insufficient.

To address this, a dynamic mask was introduced based on the solar ramp, defined as the derivative of the solar curve or the change in generation between two consecutive 30-minute periods. A ramp threshold of 0.2 MW was applied, meaning only periods where the ramp indicated active generation was occurring were included in the analysis. This ensured that error statistics were calculated only over meaningful trading periods, yielding more accurate and representative results across all auction stages.

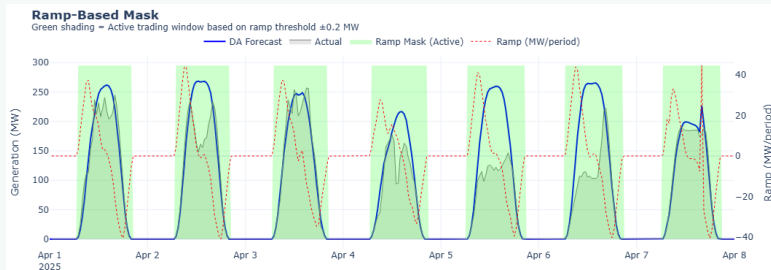
Filtering the Forecasts

- At first, all periods were being analysed (including night), which caused in the error statistics to be inaccurate.
- To improve accuracy, a static mask was used to try and filter out the periods of no solar generation.
- However, due to the solar path changing throughout the year, the periods where solar generation equals 0 MW varies.
- In order to handle this, a dynamic mask approach was used.
- The solar ramp was defined as being the difference in generation between two consecutive 30-min periods (or the derivative of the solar curve).
- A ramp threshold was set to 0.2 MW.
- This meant that only periods where solar generation was occurring were to be analysed, ultimately yielding more accurate results.

Auction:	Static Mask Definition:
DA & IDA1 Mask:	06:00 – 23:00
IDA2 Mask	11:00 – 23:00
IDA3 Mask	17:00 – 23:00

$$\text{Ramp} = \frac{d}{dt}(\text{Solar Curve})$$

Auction:	Dynamic Mask Definition:
DA & IDA1 Mask:	[0.2 MW < Ramp < 0.2 MW]
IDA2 Mask	11:00 – [Ramp < 0.2 MW]
IDA3 Mask	17:00 – [Ramp < 0.2 MW]



Forecast accuracy metrics for Forecast A were calculated across each auction stage using three masking approaches — no filter, a static time-based mask, and a dynamic ramp-based mask — to illustrate the impact of each on the reported statistics. The ramp-based mask, which restricted analysis to periods of active generation, produced notably improved results and was more representative of true forecast performance.

Forecast A Accuracy

- The expected outcome for the error distribution would logically be DA to IDA3, least to most accurate as we get closer to delivery.
- However this was not the case, as seen in the *Error (All Hours)* table. The error data was being flattened by the 0 and NaN periods within the data outside of the solar hours.
- When the static mask was applied, the distribution became more aligned with what was expected.
- When the dynamic mask was applied, the results became even more accurate. This led to the adjusted error results being much more in-line with expectation.
- A positive mean error across DA, IDA1, and IDA2 suggest we are usually long in these auctions, compared to being slightly short in IDA3.
- IDA3 was by far the most accurate forecast, despite having a high MAPE (likely caused by small division, yielding high numbers).
- IDA2 had the highest MAE and RMSE, likely due to it overcompensating for volatility.

Error (All Hours)

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	2.74	10.82	22.97	71.1%
IDA1	2.34	10.6	22.65	69.7%
IDA2	2.83	15.5	27.55	62.0%
IDA3	-0.42	4	10.08	96.2%

Adjusted Error (Static Mask)

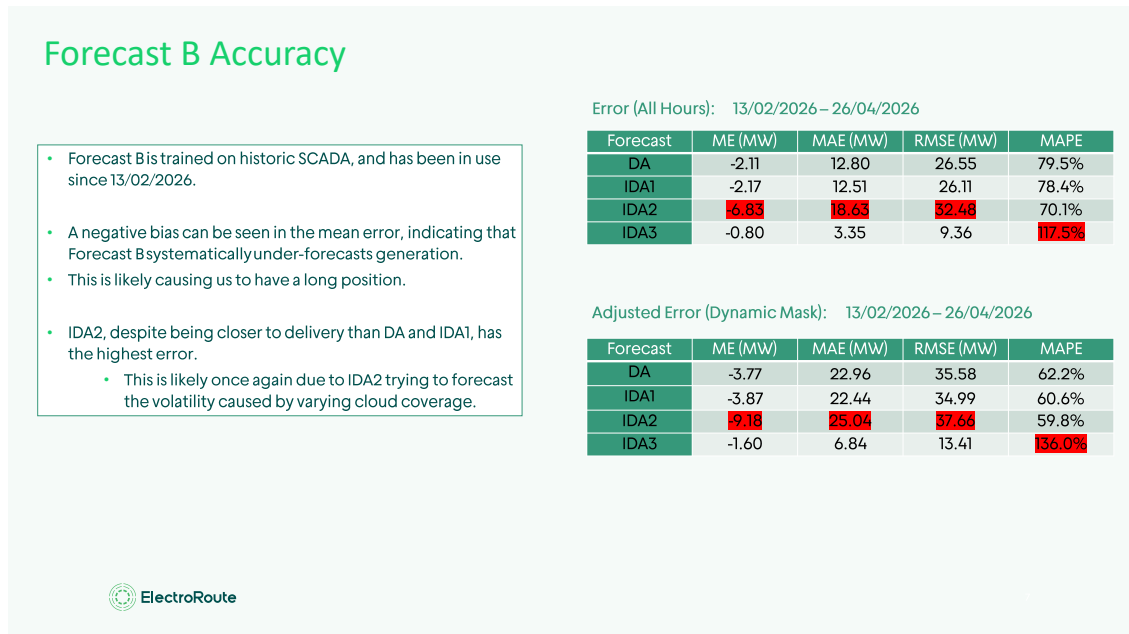
Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	3.86	15.25	27.29	63.6%
IDA1	3.31	14.95	26.91	61.7%
IDA2	2.83	15.50	27.55	62.0%
IDA3	-0.02	3.10	8.20	99.0%

Adjusted Error (Dynamic Mask)

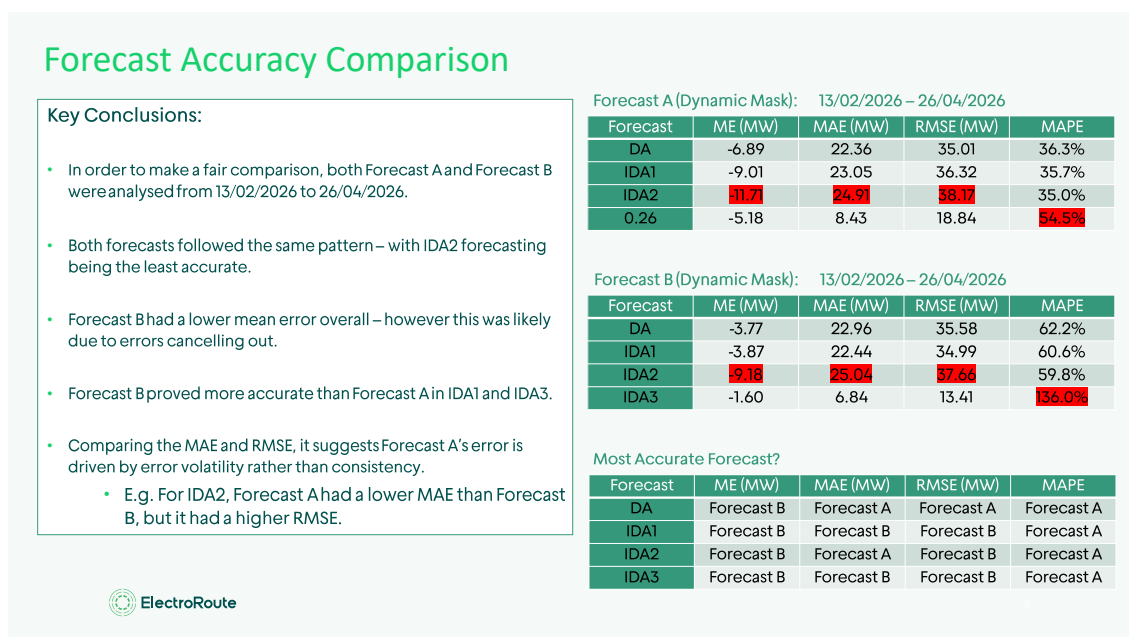
Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	5.07	19.95	31.21	57.5%
IDA1	4.34	19.55	30.77	55.4%
IDA2	3.85	21.08	32.13	53.5%
IDA3	-0.88	8.47	14.68	93.4%

4.2 Forecast Performance Analysis

Similarly, the same metrics were used for the analysis of the second forecast provider. This forecast has been in use since 13/02/2026 therefore the analysis took place over a smaller window.

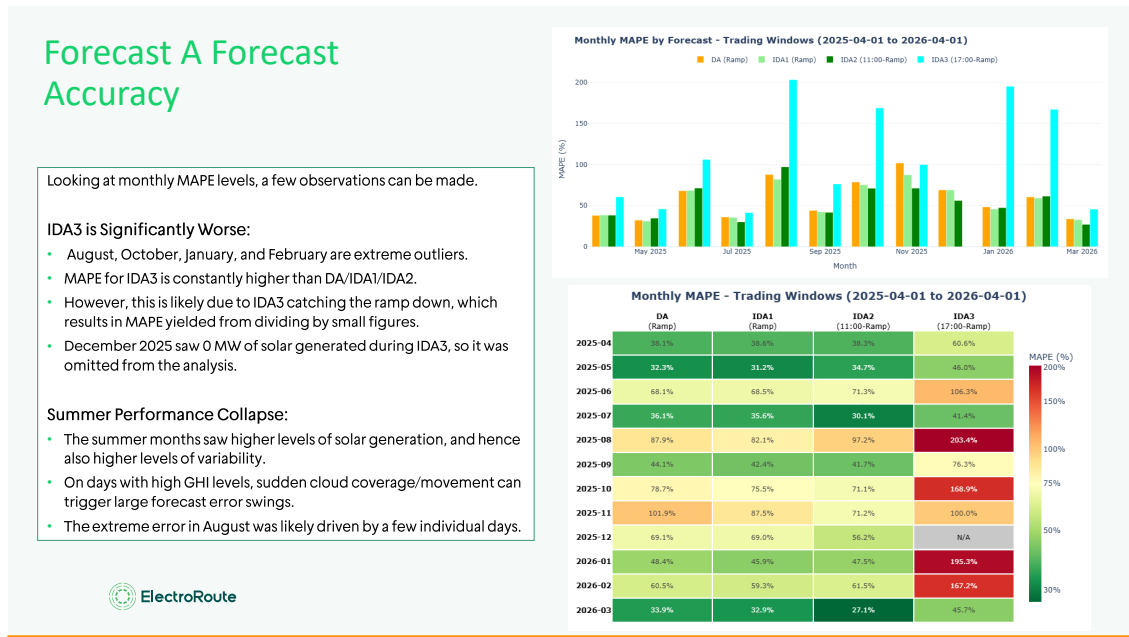


To ensure a fair comparison, both forecasts were evaluated over the same time period using the dynamic mask. Both followed the same pattern, with IDA2 being the least accurate auction stage. While Forecast B exhibited a lower mean error overall, this was likely due to errors cancelling out, with Forecast A proving more accurate across several key metrics.

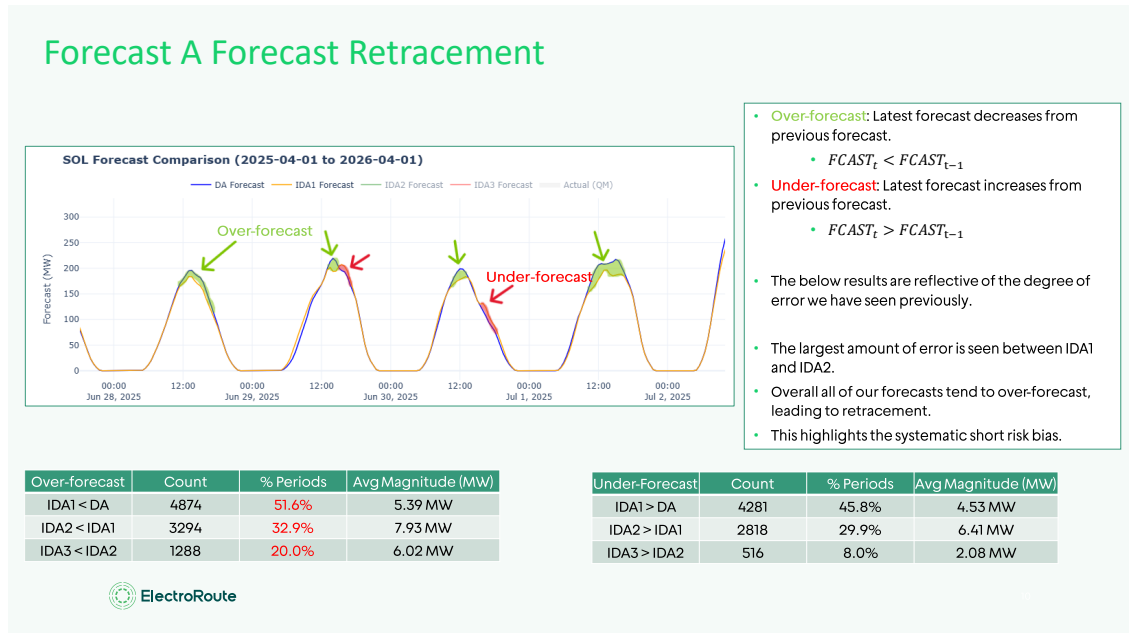


This slide presents the monthly MAPE breakdown for Forecast A across each auction stage, displayed as a bar chart and heatmap covering April 2025 to April 2026. The dynamic ramp-based mask is applied throughout, with December 2025 omitted from IDA3 as no solar generation was recorded during that period. The heatmap allows patterns in forecast accuracy to be identified across months and auction stages, with

colour scaling used to highlight periods of particularly high error.



In order to determine whether our forecasts were causing us to take short or long positions, the forecast retracement was analysed. Overall, the analysis suggested that our forecasts were over-forecasting, possibly driving the short risk bias across the portfolio.



4.3 Individual Solar Site Performance Analysis

The individual performance of each solar farm was analysed to try and identify if the errors were being driven by any particular sites. The site names and capacities were kept hidden. Scatter plots and Python's pvlib library helped us identify a systematic shortness and higher error probability during high solar days. We also identified some outliers that were skewing the results slightly.

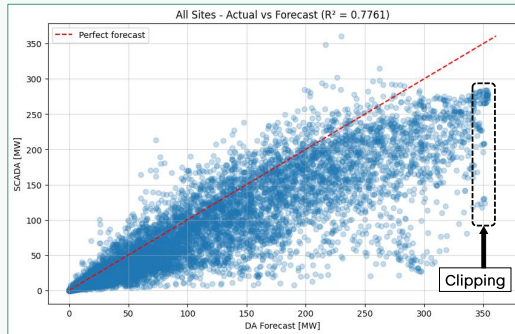
Forecast A Site Forecast Accuracy

This section of analysis looks at the SCADA / DA forecast relationship at each solar site in our portfolio.

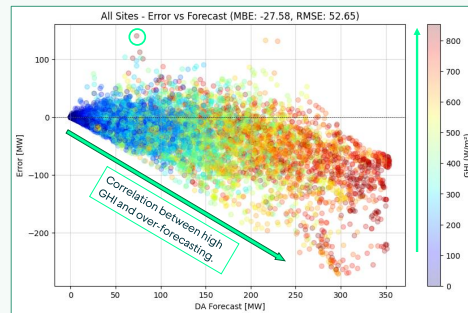
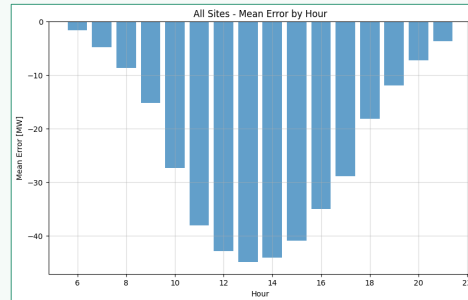
Python's pvlb library was used to model clear sky GHI output at an aggregated coordinate of all our sites.

A mask was applied to try and filter out inconsistencies:

06:30 – 22:00, excluding any periods where either SCADA, forecast, or both are NaN/0.



It is worth noting that the portfolio's maximum export capacity (MEC) is 356.4MW, which leads to a clipping effect as outlined in the plot above.



Forecast A Site Forecast Accuracy

'n' represents the amount of samples that were used in the analysis, with the red cells indicating significant data exclusion (mostly due to consistent SCADA outages).

Portfolio Metric	Value	%
ME	-27.58 MW	-7.64%
MAE	33.53 MW	9.28%
RMSE	52.65 MW	14.58%
R²	0.7761	-
Over-Forecast	-	80.4%
Under-Forecast	-	19.6%

Per-Site Forecast Error Statistics (sorted by nMAE)

Site	Capacity (MW)	n	ME (MW)	MAE (MW)	RMSE (MW)	nME (%)	nMAE (%)	nRMSE (%)	R²	Over (%)	Under (%)
		3029	0.02	0.44	0.72	0.25	4.86	8.03	0.68	49.72	50.28
		8414	-0.12	0.63	0.95	-1.43	7.6	11.53	0.74	56.23	43.77
		3289	0.12	0.4	0.61	2.47	8.13	12.24	0.82	44.6	55.4
		7873	-1.58	5.13	8.12	-2.64	8.55	13.53	0.77	59.75	40.25
		8496	-0.15	0.43	0.67	-3.07	9.11	14.12	0.71	59.52	40.35
		3294	-0.05	0.45	0.72	-0.99	9.16	14.45	0.71	55.49	44.47
		3296	0	0.46	0.69	0.01	9.25	13.85	0.72	52.12	47.88
		7450	-0.49	2.12	3.23	-2.26	9.73	14.86	0.68	56.27	43.73
		3307	0.1	0.57	0.88	2	11.46	17.85	0.53	44.84	55.16
		6321	-2.21	10.37	15.9	-2.54	11.91	18.28	0.66	56.97	42.97
		8262	-7.58	14.31	21.85	-6.37	12.02	18.36	0.65	64.43	35.57
		5498	-3.25	4.61	7.35	-10.29	14.59	23.25	0.38	71.12	28.88

Forecast A Site Accuracy

Looking at each Irish solar site individually, we can see how each one contributes to the overall forecast error.

The site error distribution is skewed negatively with the larger sites positioned toward the upper tail. However there are some outliers such as [SITE NAME] which is a relatively large site with decent accuracy.

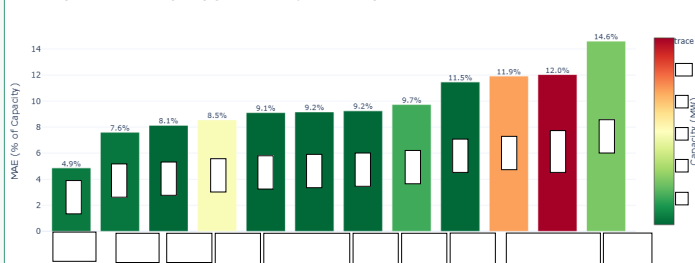
For values where the MBE is negative, the SCADA is less than the forecasted volume (site over-forecasting). This means that we are likely to be short if trading to forecast.

On average, the sites over-forecasted at all but two sites, meaning our solar book was more likely to be short across the last year. Therefore there appears to be a systematic short risk across our portfolio.

It is worth mentioning that this analysis is based on SCADA, which excludes a lot of null periods. Further analysis will use metered generation.



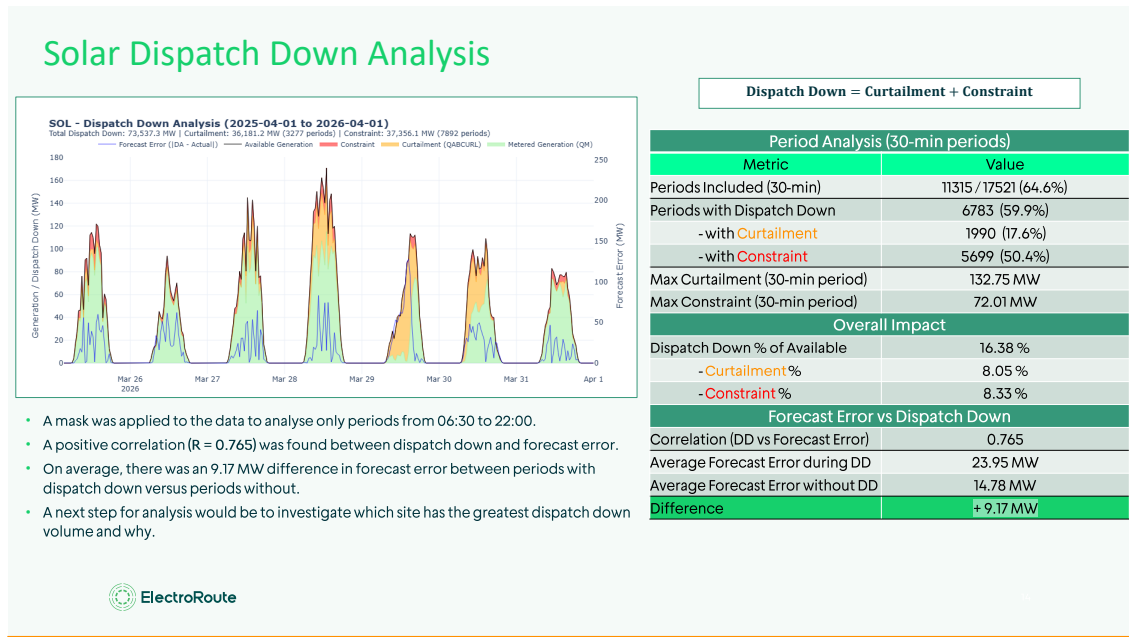
MAE by Site as % of Capacity (06:30-22:00, excl. zeros)



MBE by Site as % of Capacity (06:30-22:00, excl. zeros)

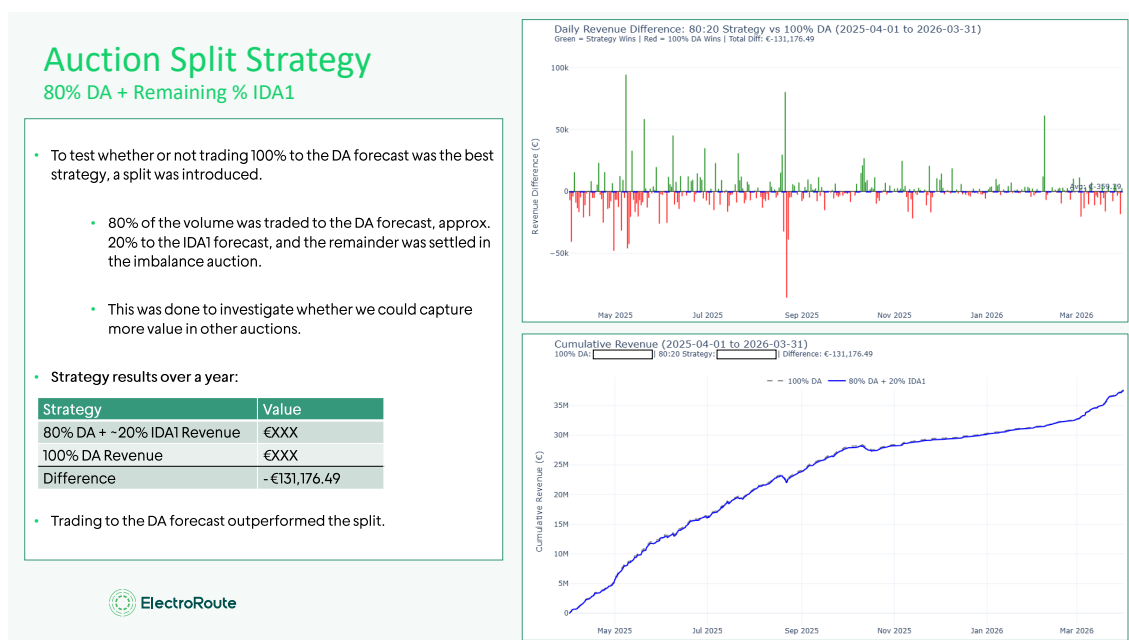


Analysis was also done to investigate whether or not there was a correlation between dispatch down and forecast error. A positive correlation was found between the two, with an overall difference of 9.17 MW between error during dispatch down and without.



4.4 Auction Split Strategy Backtest

The following slides look at some backtested strategies. This was less of a data-driven analysis, but rather just an investigation into whether modifying how we traded our volume would have an effect on performance, examining an 80:20 auction split. Rather than trading 100% of volume to the DA forecast, 80% was traded to DA, approximately 20% to IDA1 or IDA2, with the remainder settled in the imbalance auction. Revenue and consistency were compared with a 100% baseline.

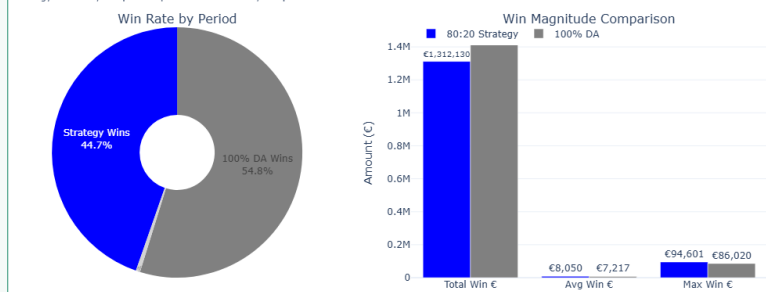


Split Analysis

80% DA + Remaining % IDA1

- A comparison of each individual traded period can tell us whether the DA strategy won through consistent smaller wins or occasional large wins.
- 100% DA had a better win rate.
- 100% DA had a greater win total.
- 80/20% IDA1 had a greater average win.
- 80/20% IDA1 also had a greater max win.
- This suggests that the 80/20% split is more volatile but is not as consistent as 100% DA.
- Overall trading 100% DA was better.

Big Wins vs Consistency Analysis (2025-04-01 to 2026-03-31)
Strategy wins 163/365 periods | 100% DA wins 200/365 periods



Metric	100% DA	80/20%
Win Rate	54.8%	44.7%
Total Win	€1,443,306	€1,312,130
Average Win	€7,217	€8,050
Maximum Win	€86,020	€94,601



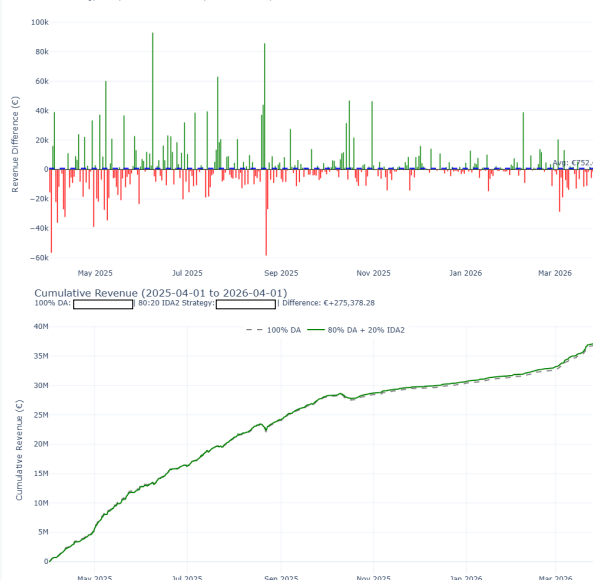
Split Analysis

80% DA + Remaining % IDA2

- The same test was done with IDA2 instead of IDA1.
 - 80% of the volume was traded to the DA forecast, approx. 20% to the IDA2 forecast, and the remainder was settled in the imbalance auction.
 - This was done to investigate whether we could capture more value in other auctions.
- Strategy results over a year:

Strategy	Value
80% DA + ~20% IDA2 Revenue	€XXX
100% DA Revenue	€XXX
Difference	+ €275,378.28
- The IDA2 split outperformed trading 100% to DA.

Daily Revenue Difference: 80:20 IDA2 Strategy vs 100% DA (2025-04-01 to 2026-04-01)
Green = Strategy Wins | Red = 100% DA Wins | Total Diff: €+275,378.28

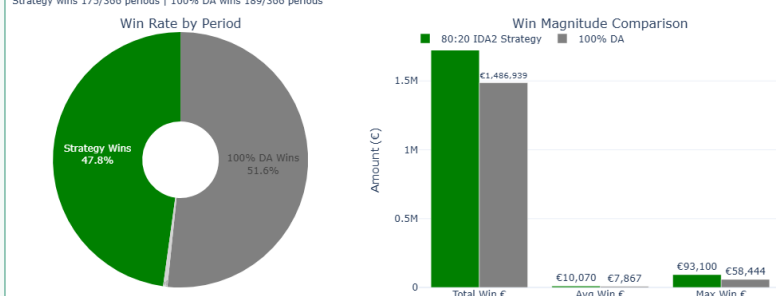


Split Analysis

80% DA + Remaining % IDA2

- 100% DA had a better win rate.
- 80/20% IDA2 had a greater win total.
- 80/20% IDA2 had a greater average win.
- 80/20% IDA2 also had a greater max win.
- This suggests that the 80/20% split is more volatile but is not as consistent as 100% DA.
- The IDA2 split performed better than 100% DA.
- The IDA2 split outperformed the IDA1 split.

Big Wins vs Consistency Analysis (2025-04-01 to 2026-04-01)
Strategy wins 175/366 periods | 100% DA wins 189/366 periods

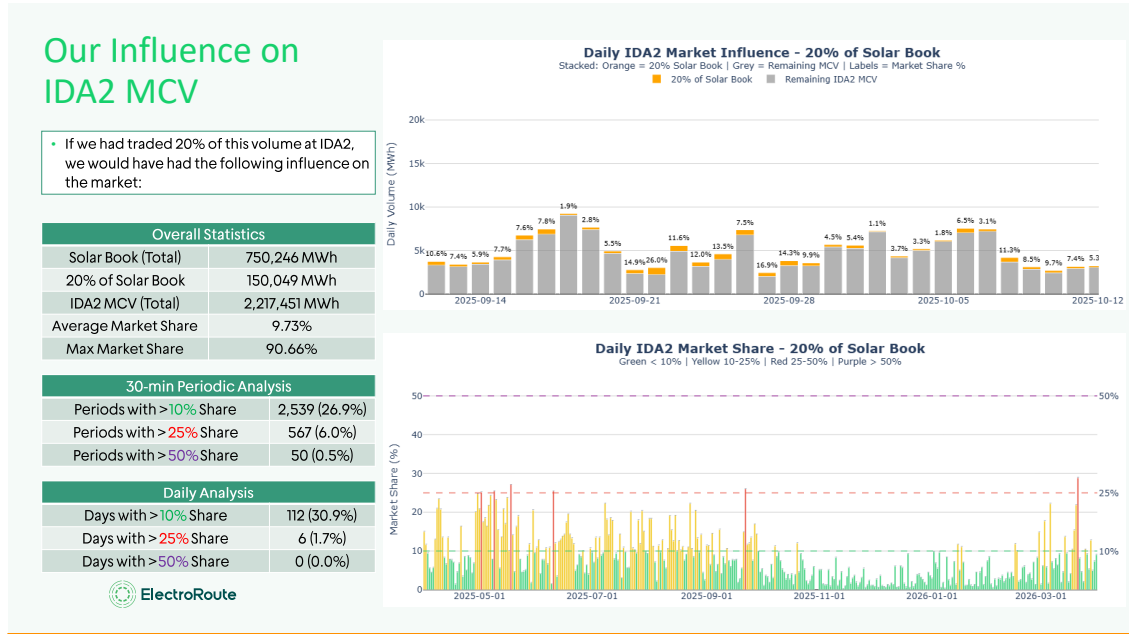


Metric	80/20% IDA1	80/20% IDA2
Total Revenue	€XXX	€XXX
Difference		+ €443,444.43

Metric	100% DA	80/20%
Win Rate	51.6%	47.8%
Total Win	€1,486,939	€1,762,317
Average Win	€7,867	€10,070
Maximum Win	€58,444	€93,100

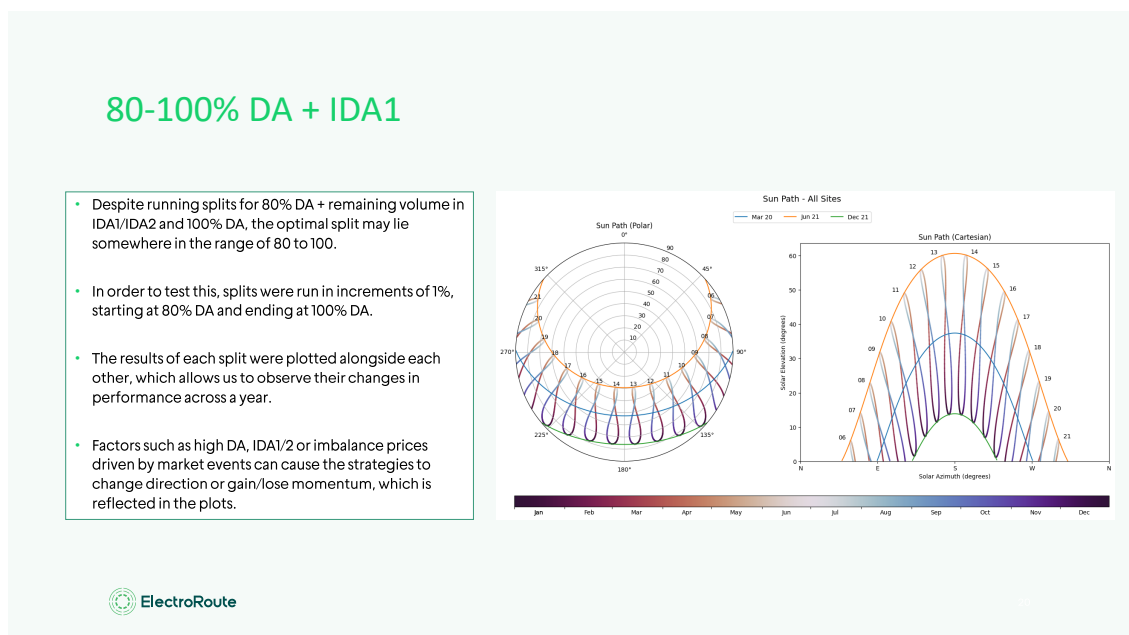


In order to determine if trading 20% of our volume in IDA2, we looked at the MCV and how we would influence it. Across all 30-minute trading periods, we held approximately 10% of the market clearing volume. This is not very desirable as it means our volume has the power to move prices and the system imbalance, which could lead to unexpected changes.



4.5 Finding the Optimal Split

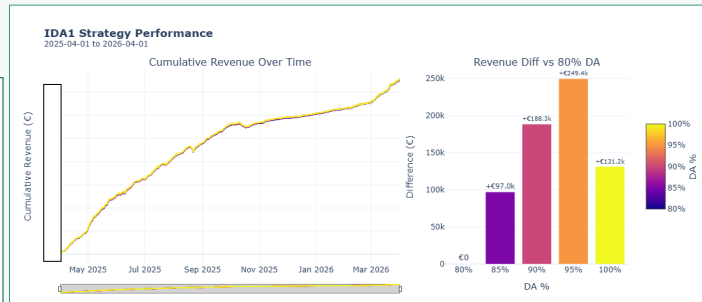
This section explores the range between the 80% DA and 100% DA strategies, acknowledging that the optimal split may lie somewhere between the two extremes. To investigate this, splits were run in increments of 1% from 80% DA to 100% DA, with the remaining volume traded in IDA1. The same approach was also applied for an IDA2 split, running the same incremental analysis with the remaining volume traded in IDA2 instead.



80-100% DA + IDA1

- Splitting over DA and IDA1 produced fairly consistent results. Across the year, lower splits (purple) performed worse.
- The higher splits consistently performed better with the upper-middle splits generating the most revenue from November 2025 onward.
- Overall, the best split was 98% DA.
- The results suggest that there is value in holding back some volume to trade in IDA1.

Metric	Strategy	Revenue
Best IDA1 Strategy	98% DA	€XXX
Baseline Strategy	100% DA	€XXX
Difference		€133,575.61

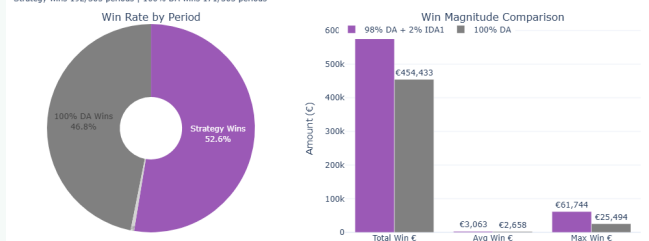


Optimal IDA1 Split 98% DA + Remaining % IDA1

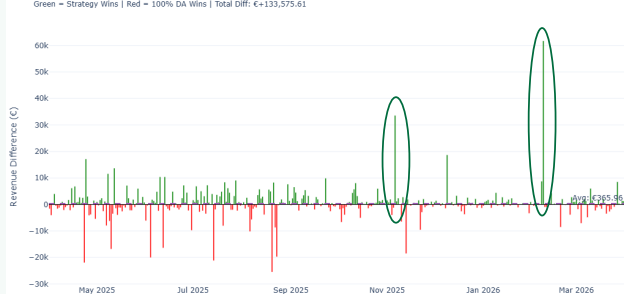
- The strategy won 52.6% of the time compared to 100% DA.
- It also had a greater total win, average win, and maximum win.
- However it did have two days of massive wins, outlined in the figure to the right, that could have driven the result.
- But overall it was more consistent too.

Metric	100% DA	98/2%
Win Rate	46.8%	52.6%
Total Win	€454,433	€588,008
Average Win	€2,658	€3,063
Maximum Win	€25,494	€61,744

Big Wins vs Consistency Analysis (2025-04-01 to 2026-03-31)
Strategy wins 192/365 periods | 100% DA wins 171/365 periods



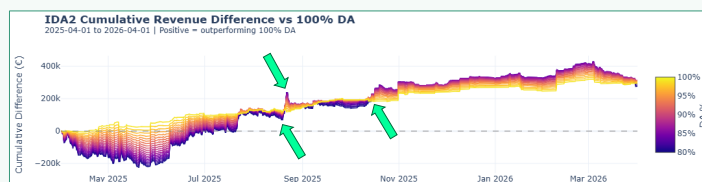
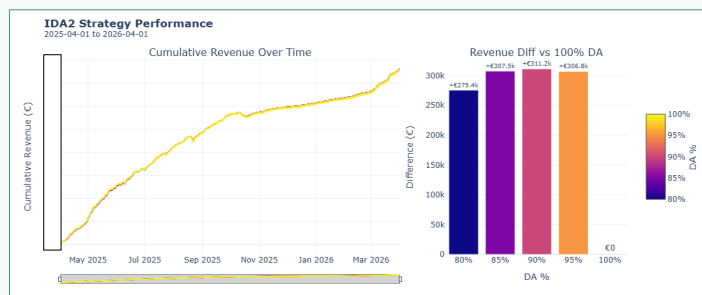
Daily Revenue Difference: 98% DA + 2% IDA1 Strategy vs 100% DA (2025-04-01 to 2026-03-31)
Green = Strategy Wins | Red = 100% DA Wins | Total Diff: €+133,575.61



80-100% DA + IDA2

- The expected price volatility of IDA2 can be seen here, especially on August 21st and 22nd where high forecast error caused significant gains and losses.
- This volatility also resulted in multiple inflection points to be generated along the curve, highlighted by the arrows below.
- Overall, the IDA2 strategy outperformed the IDA1 strategy, with both the best and worst IDA2 strats outperforming the best and worst IDA1 strats.
- However, the IDA2 strategy yielded greater dispersion, which once again reflects this higher volatility and greater risk.

Metric	Strategy	Revenue
Best IDA2 Strategy	90% DA	€XXX
Baseline Strategy	100% DA	€XXX
Difference		€311,207.81

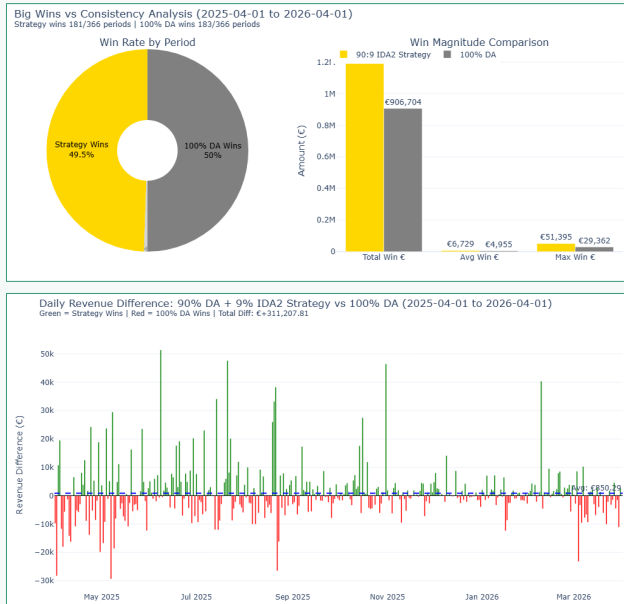


Optimal IDA2 Split

90% DA + Remaining % IDA2

- In terms of consistency, the 100% DA split outperformed the 90% split slightly.
- The 90% split had a greater total win, average win, and maximum win.
- From the graph below it looks like there were multiple high magnitude revenue days driving the performance.

Metric	100% DA	98/2%
Win Rate	50%	49.5%
Total Win	€906,704	€1,217,911
Average Win	€4,955	€6,729
Maximum Win	€29,362	€51,395



Trading 10% of the volume at IDA2 yielded the best returns. Therefore, the % of MCV was investigated. Overall it had a moderate but manageable influence on the MCV — making it unlikely to influence prices too much.

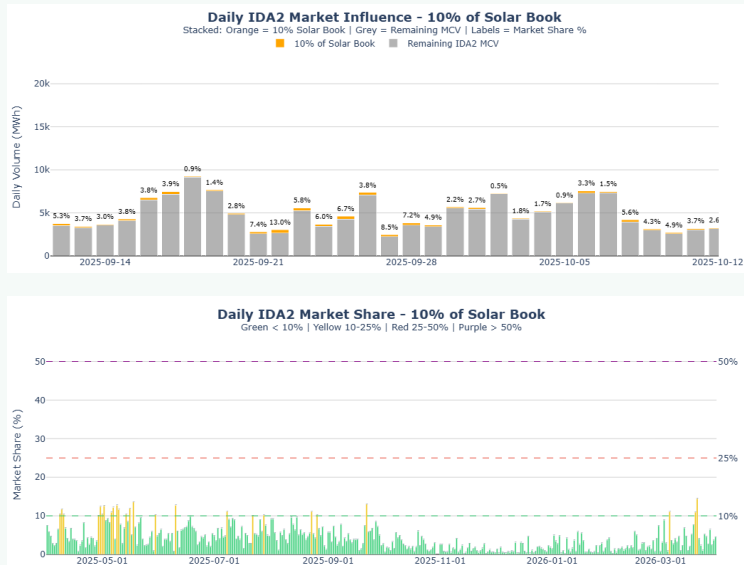
Our Influence on IDA2 MCV

- If we had traded 10% of this volume at IDA2, we would have had the following influence on the market:

Overall Statistics	
Solar Book (Total)	750,246 MWh
20% of Solar Book	75,025 MWh
IDA2 MCV (Total)	2,217,451 MWh
Average Market Share	4.86%
Max Market Share	45.33%

30-min Periodic Analysis	
Periods with >10% Share	949 (10.1%)
Periods with >25% Share	50 (0.5%)
Periods with >50% Share	0 (0.0%)

Daily Analysis	
Days with >10% Share	25 (6.9%)
Days with >25% Share	0 (0.0%)
Days with >50% Share	0 (0.0%)

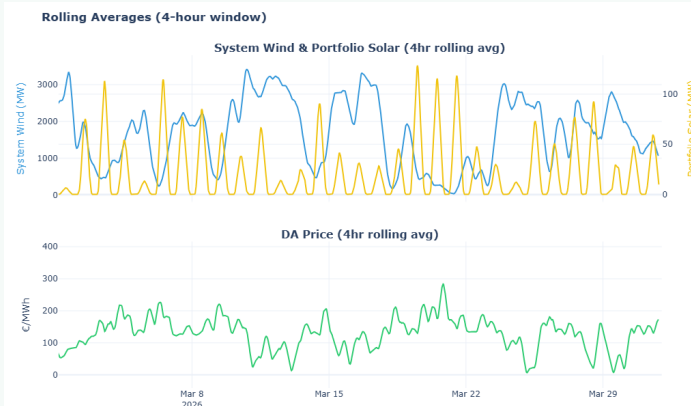


4.6 Influence of Wind and Dispatch Down

The relationship between wind generation and Day-Ahead (DA) prices was investigated, revealing a moderate negative correlation between the two, with high wind periods suppressing prices and impacting the profitability of solar trading. A case study of August 2025 was presented to illustrate this, with wind levels categorised into three groups to highlight the role of seasonality in trading performance. Finally, solar and wind curtailment were examined, with the vast majority of solar curtailment found to occur during periods of concurrent wind curtailment, suggesting wind curtailment is a strong predictor of solar curtailment.

Influence of Wind on Prices

- We know that high wind tends to push prices down, but here are some stats that back this up.



Key Statistics:

Wind vs DA Price Correlation **-0.454**

- Moderate negative relationship. As wind increases, DA price tends to decrease.

Wind-Price Sensitivity **-0.025 €/MW**

- For every 1 MW increase in system wind, the DA price drops by €0.025 (-€25.08 per 1000 MW).

R-Squared (R^2) **0.206**

- Roughly 21% of DA price variance is driven by wind alone. Other factors such as demand, solar, gas prices etc. drive the remaining 79%.

P-Value (P) **~0.00**

- The relationship is statistically significant and is not based on random chance.



Influence of Wind

Example: August 2025

- Here is plot for August 2025, a particularly high solar month where we had large balancing costs.
- Wind levels were low, solar levels were high, and prices remained relatively stable across DA, IDA1 & IDA2. However the imbalance prices were very volatile.
- When wind came up during these periods prices were pushed down, with imbalance prices being completely floored in some instances.
- The opposite is also true, low wind and high solar was extremely volatile in the BM.
- It is clear that seasonality plays a significant role in determining how effectively we can trade solar (i.e. large imbalances in the summer have greater impact).
- Further analysis is required to investigate why solar is being undersold on low wind days—specifically whether this is driven by wind forecast error or an expectedly low wind output.
- Based on percentages of the maximum, wind levels were categorised into 'High' (>66%), 'Medium' (33-66%), and 'Low' (<33%).

FY25/26

Wind	Periods	Wind (MW)	DA (€)	IDA1 (€)	IDA2 (€)	Imbalance (€)	DA vs Low Wind
Low	2940	344	140.1	138.95	138.36	152.11	-
Medium	2832	1150	104.5	101.94	98.91	100.39	-35.59
High	2581	2365	85.28	84.19	80.29	78.38	-54.81

Aug 25

Wind	Periods	Wind (MW)	DA (€)	IDA1 (€)	IDA2 (€)	Imbalance (€)	DA vs Low Wind
Low	256	181	135.39	128.48	134.99	181.99	-
Medium	308	792	96.63	96.82	91.31	109.51	-38.76
High	269	1,908	69.32	69.00	66.10	55.94	-66.08



Curtailment Analysis

- It is worth investigating whether there is any correlation between solar and wind curtailment.

Category	Count	Percentage
Total 30-min Periods	17,518	100%
Neither Curtailed	13,196	77.1%
Wind Only Curtailed	1,881	10.7%
Solar Only Curtailed	163	0.9%
Both Curtailed	1,963	11.2%

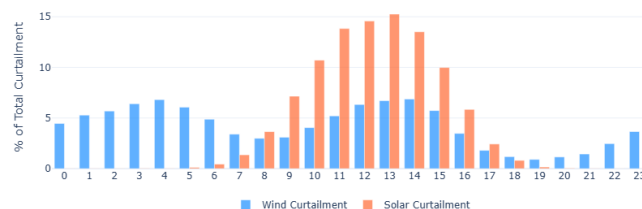
Metric	Value (MW)	Percentage
Total Wind Curtailment	68,386.4	65.4% of total
Total Solar Curtailment	36,181.2	34.6% of total
Wind During Concurrent	40,583.0	59.3% of wind
Solar During Concurrent	35,617.0	98.4% of solar

- Based on this analysis, it is fair to say that there is a relationship between solar and wind curtailment – and therefore forecasting wind curtailment could possibly help us trade solar more effectively.
- Further analysis could include studying the impact of over selling solar on high wind days.

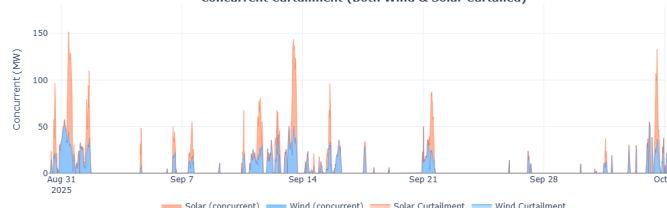


Curtailment Distribution by Hour of Day

% of total curtailment occurring in each hour



Concurrent Curtailment (Both Wind & Solar Curtailed)



The analysis concluded that there were multiple direct and indirect factors influencing solar book performance. Forecast accuracy was identified as the most influential factor, with a systematic over-forecasting bias causing consistent short positions. Curtailment created significant unavoidable exposure, particularly during peak solar hours, with further site-level analysis required to understand the magnitude of errors at individual sites. Wind was found to be the key driver of Irish electricity prices, with high wind periods correlating with both lower prices and increased curtailment risk, making wind forecasts essential for optimising the solar trading strategy. Finally, trading activity had the potential to influence prices and imbalance volumes depending on the auction in which volume was traded, which had to be carefully managed. Addressing forecast bias and understanding site-level performance variations represented the highest-impact opportunities for improving solar book profitability.

Conclusions

Based on this analysis, we can conclude that there are multiple different factors that direct and indirect influences on our solar book performance.

- **Forecast Accuracy** is without doubt the most influential factor on our performance. From this analysis, we have noticed a systematic over-forecasting bias – which consequently has caused us to be short.
- **Curtailment** creates significant unavoidable exposure particularly during peak solar hours.
 - Further analysis needs to be done on the individual sites such as [SITE NAME] and [SITE NAME] to determine why the magnitude of over-forecasting and error are so large.
- **Wind** is the key driver of Irish electricity prices. High wind periods correlate with both lower prices and increased solar curtailment risk, making wind forecasts essential for optimising our solar trading strategy.
- Depending on which auction we trade our volume in, we can have a significant influence on the **Market Clearing Volume**. This means that our trading activity has the potential to influence prices and imbalance volumes – which is something we need to be careful of.
 - A practical example of this was simulating 20% of our solar volume through IDA2, representing 10% of the total MCV.

Addressing forecast bias and understanding site-level performance variations represent the highest-impact opportunities for improving our solar book profitability.



4.7 Next Steps

The next steps outlined focused on strategy development and testing, with the aim of optimising volume in the IDAs to reduce balancing exposure. A scoring system was proposed to categorise the forecast on a given day based on volume levels, peak level and ramp rate, with seasonality accounted for using a Normalised Solar Index (NSI). The NSI was defined as the daily total forecasted volume divided by the 14-day rolling median, providing a smoother and more reliable measure of seasonality than raw forecasts alone.

To determine the optimal auction split for a given day, an ensemble model approach was proposed, whereby multiple models and their weights were learned automatically from a set of feature inputs. These inputs included solar and wind generation forecasts, system demand forecasts, historical forecast error metrics, auction price history, price spreads, historical price volatility, imbalance price flex, SNSP forecasts, curtailment events and expected ramp rates. The weighted ensemble would then output a recommended split for the following trading day. This was framed as a revenue maximisation problem, where predicting the optimal auction split for a given day would allow trading decisions to be made that maximise revenue. It was noted that this approach could be implemented with as little as one year of data, with model weights updated dynamically each trading day.

Next Steps

Strategy Development and Testing

- Strategy Development Aim – optimise volume in the IDA's to reduce balancing exposure.
- How - To optimise the volume between auctions, proposing to use a scoring system to categorise the forecast on a given day.
- The system looks to categorise the forecast based on volume levels, peak level and ramp rate. Seasonality is reduced using NSI (normalised solar index Forecast/14day rolling median - in winter you can extend the formula to include seasonal average using 2/3 years of data).
- Backtest the scores with associated % split to see the most profitable (if that is true at all)
- If it's true, shadow trade.
- Investigate underselling solar on low wind days and adding as a screen to the scoring

Date	Month	Forecast MWh	14d Median Expected	NSI	NSI Bucket	Peak MW	Peak	Peak Bucket	Max Ramp MW	Ramp	Ramp Bucket	Score	DA Sell %	IDA 2 Sell %
01-Apr	Apr	95	118	0.81	Medium	19	0.63	Medium	4	0.13	Low	4	95.0%	5.0%
02-Apr	Apr	140	120	1.17	High	27	0.90	Medium	8	0.27	Medium	4	95.0%	5.0%
03-Apr	Apr	155	122	1.27	High	29	0.97	High	6	0.20	Medium	5	97.5%	2.5%
04-Apr	Apr	50	121	0.41	Low	15	0.50	Low	5	0.17	Medium	1	85.0%	15.0%
05-Apr	Apr	180	125	1.44	High	30	1.00	High	9	0.30	High	4	95.0%	5.0%
06-Apr	Apr	125	128	0.98	Medium	24	0.80	Medium	5	0.17	Medium	3	92.5%	7.5%

NSI	NSI = Forecast Daily MWh / 14-day rolling median expected MWh.
Peak	Peak = Forecast Peak MW / Installed Capacity. For a 30 MW site, Peak = 1.00 means the forecast reaches full output.
Ramp	Ramp = Max half-hour MW change / Installed Capacity. For a 30 MW site, Ramp = 0.33 means the largest half-hour forecast step is 10 MW.
Scoring	This example uses a simple additive score: NSI Low/Medium/High = 0/1/2; Peak Low/Medium/High = 0/1/2; RampAdj Low/Medium/High = 2/1/0. Lower ramp gets a higher score because smoother days are typically easier forecast



Normalised Solar Index (NSI) - Example

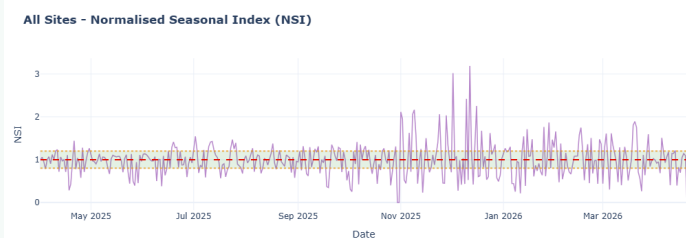
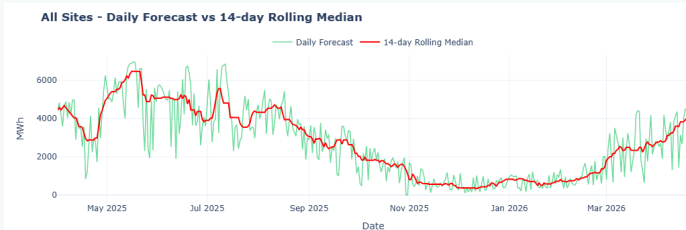
- A rolling median can also help us determine the seasonality of solar generation by creating a smoother curve.
- However this doesn't necessarily give us a good edge to predict the solar output due to its naturally high daily volatility – particularly in the summer months as seen on the right.
- A normalised solar index was defined as:

$$NSI = \frac{\text{Daily Total Forecasted Volume}}{14 - \text{Day Rolling Median}}$$

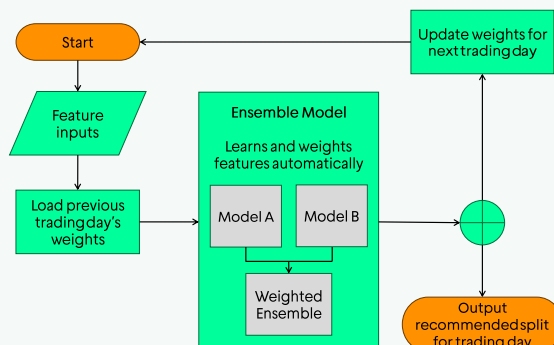
NSI Value	Meaning
NSI = 1.0	Exactly average day (matches rolling median)
NSI > 1.0	Above-average solar day (better than recent)
NSI < 1.0	Below-average solar day (worse than recent)
NSI > 1.2	Exceptionally good day (20% above recent)
NSI < 0.8	Exceptionally poor day (20% below recent)

- Results:

Metric	Value
Average NSI	0.997
NSI Standard Deviation	0.402
Days Above Normal (> 1.2)	93 (24.2%)
Days Below Normal (< 0.8)	107 (27.8%)
Days Normal (0.8 – 1.2)	185 (48.1%)



How Could This Be Done?



Example Feature Inputs	Weights
Solar Generation Forecasts (DA, IDA1, IDA2)	...
Wind Generation Forecasts (DA, IDA1, IDA2)	...
System Demand Forecast	...
Historical Forecast Error Metrics (ME, MAE, RMSE)	...
Forecast Bias History (systematic over/under-forecasting)	...
Auction Price History (DA, IDA1, IDA2, Imbalance)	...
Price Spreads (DA-IDA1, IDA1-IDA2, IDA2-Imbalance)	...
Historical Price Volatility by Auction	...
Imbalance Price Bias (long vs short penalty)	...
SNSP Forecast	...
Curtailment Events	...
Historical MCV %	...
Hour of Day & Day Type (Weekday/Weekend)	...
Season / Time of Year	...
Expected Ramp Rate (MW/hour)	...
Sum	1.00

- This is a revenue maximisation problem that can be resolved by predicting what the optimal auction split for a given day will be.
- By using some of the data features we have just analysed, alongside our forecasts, a recommendation can be made for the following trading day on what the best volume split would be to maximise revenue.
- This could be done by using an ensemble or some form of regression, where each feature is an input to the model and can be dynamically weighted from 0-1, depending on how much influence it would have on a particular day.

4.8 Skills and Learning Outcomes

Completing this analysis developed a range of both technical and professional skills. From a technical perspective, the project required extensive use of Python and SQL to extract, clean and process large volumes of market data across multiple auction stages. Writing complex queries to retrieve data from internal databases and manipulating it in Python using libraries such as Pandas, Plotly, and even learning new libraries such as pvlib, strengthened my ability to work efficiently with real-world datasets. The importance of ensuring data is cleaned effectively before analysing it was something that stood out to me. The analysis also provided valuable insight into the factors driving solar book performance, including forecast accuracy, curtailment and wind, which gave me a much stronger understanding of how renewable assets are traded in the Irish electricity market.

Presenting the findings to the full trading team was a particularly valuable experience. Communicating technical results clearly and concisely to a professional audience required careful consideration of how to structure and deliver the material. Fielding questions from the team further developed my ability to think critically under pressure and defend my methodology and conclusions. Overall, this project significantly strengthened my confidence in both conducting independent analysis and presenting findings in a professional setting.

5 Machine Learning Demand Forecasting Ensemble

Another project undertaken during my placement involved the development of a machine learning ensemble model to more accurately forecast system-wide electricity demand. ElectroRoute currently uses multiple demand forecasts, however these rarely agree with one another, creating uncertainty when making trading decisions.

The project began as a straightforward analysis requested by one of the traders, who wanted to investigate the difference in error between the two primary forecasts used by the team. The first is provided by the TSO, EirGrid, and the second by an external forecast provider, whose name has been redacted from this document for confidentiality reasons. Upon completing the initial analysis, it became clear that each forecast had distinct strengths and weaknesses. The EirGrid forecast consistently performed better at predicting the troughs of the demand curve, while the external provider was more accurate in capturing peak demand periods. Recognising this complementary behaviour, the decision was made to take the initiative and attempt to develop a blended ensemble model that could leverage the strengths of both forecasts, producing a more accurate and robust prediction of system demand.

5.1 Model Selection

For the ensemble model, ridge regression was selected as the most appropriate modelling approach. Ridge regression is a form of linear regression that introduces a regularisation term, penalising large coefficients and reducing the risk of overfitting. Rather than treating the two forecasts equally, the model learns an optimal weighting for each based on historical performance, allowing it to allocate more weight to whichever forecast has proven more accurate. Given the relatively small number of input features and the need for a transparent and explainable model that could be presented to the trading team, ridge regression struck the right balance between simplicity, robustness and predictive performance.

Standard linear regression uses a method called ordinary least squares (OLS), which finds the coefficients that minimise the sum of squared differences between predictions and actual values. However, when input features are highly correlated—as is the case with two forecasts predicting the same quantity—OLS can produce unstable coefficient estimates that fluctuate dramatically with small changes to the training data.

Ridge regression addresses this by adding a penalty term proportional to the squared magnitude of the coefficients. The model minimises the following loss function:

$$\mathcal{L}(\beta) = \sum_{i=1}^n \left(y_i - \mathbf{x}_i^\top \beta \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \quad (1)$$

The first term measures how well the model fits the data (the standard regression objective), while the second term penalises large coefficient values. The parameter λ controls the strength of this penalty—larger values produce more conservative, stable estimates. This regularisation prevents the model from assigning extreme weights to either forecast, producing a more balanced and generalisable ensemble. Table 4 provides a summary of the notation used.

Symbol	Description
$\mathcal{L}(\beta)$	Loss function (the value to be minimised)
n	Number of observations in the training set
y_i	Actual demand value for observation i
$\mathbf{x}_i$	Input features for observation i (the two forecasts)
β	Vector of coefficients (weights) to be learned
$\mathbf{x}_i^\top \beta$	Model prediction for observation i
p	Number of predictors (in this case, 2)
β_j	Coefficient for predictor j
λ	Regularisation parameter (set to 1.0 in this implementation)

Table 4: Notation used in the ridge regression loss function.

5.2 Implementation

The ridge regression model was implemented in Python using the `scikit-learn` library. The core function, `method_ml_forecast()`, takes the combined forecast dataset as input and returns both the blended forecast and a summary of the model weights and error metrics. The key steps of the implementation are outlined below.

```
from sklearn.linear_model import Ridge
from sklearn.model_selection import train_test_split
```

Data Preparation and Train/Test Split

```
X = df[['external_forecast', 'eirgrid']].values # Input features: the two
↳ forecasts
y = df['actual'].values # Target: actual system demand

# Split into 80% training and 20% test data
# Randomised split ensures the model sees a variety of conditions in both
↳ sets,
# producing a more robust test of generalisation
X_train, X_test, y_train, y_test, idx_train, idx_test = train_test_split(
    X, y, indices,
    test_size=0.2,
    random_state=42, # Ensures reproducibility
    shuffle=True
)
```

Fitting the Ridge Regression Model

```
# Fit the ridge regression model
# alpha=1.0 controls regularisation strength - prevents overfitting
# The model learns: actual = w1*external_forecast + w2*eirgrid + intercept
# where w1 and w2 are the optimal weights for each forecast
model = Ridge(alpha=1.0)
model.fit(X_train, y_train)

# Normalise the coefficients so that the weights sum to 1
# This makes the output interpretable as a weighted blend of the two forecasts
weights = model.coef_ / model.coef_.sum() if model.coef_.sum() != 0 else
↪ model.coef_
```

Generating the Blended Forecast and Evaluating Performance

```
# Use the learned weights to generate the blended forecast across all data
df['forecast'] = model.predict(X)
df['error'] = df['forecast'] - df['actual'] # Positive = over-forecast,
↪ Negative = under-forecast

# Evaluate model performance on both training and unseen test data
weights_info = {
    'external_forecast' : weights[0],
    'eirgrid'          : weights[1],
    'intercept'         : model.intercept_,
    'train_rmse'        : np.sqrt(((model.predict(X_train) - y_train) **
    ↪ 2).mean()),
    'test_rmse'         : np.sqrt(((model.predict(X_test) - y_test) **
    ↪ 2).mean()),
    'train_mae'         : np.abs(model.predict(X_train) - y_train).mean(),
    'test_mae'          : np.abs(model.predict(X_test) - y_test).mean(),
}
```

5.3 Model Output and Results

```
=====
ML RIDGE REGRESSION
=====

Split Type: Randomised
Random State: 42
Train/Test Split: 12982 / 3246 samples
Weights: External Forecast=0.625, EirGrid=0.375
Intercept: -11.808
Train RMSE: 132.94 MW | Train MAE: 101.85 MW
Test RMSE: 134.07 MW | Test MAE: 102.66 MW
=====
```

The model was backtested over a year from 01/04/2025 to 01/04/2026. Its output revealed several interesting findings. The ridge regression model assigned a weight of 0.625 to the External Forecast and 0.375 to the EirGrid forecast, meaning the blended prediction drew more heavily from the External Forecast. This was consistent with the findings from the initial error analysis, which suggested that the External Forecast tended to perform better across a broader range of conditions, particularly during peak demand periods. The small negative intercept of -11.808 MW was a minor bias correction term learned by the model, indicating a slight systematic over-prediction in the combined forecasts that the model adjusted for.

A key indicator of model quality was the comparison between the training and test error metrics. The training RMSE of 132.94 MW and test RMSE of 134.07 MW were very close to one another, as were the training MAE of 101.85 MW and test MAE of 102.66 MW. This small gap between training and test performance was a strong sign that the model generalised well to unseen data and had not overfit to the training set. Had the model overfit, one would have expected to see a significantly lower training error relative to the test error. The consistency between the two suggested that the regularisation applied by ridge regression had been effective.

The forecast accuracy comparison tables provided a more comprehensive picture of how the ML Ridge model performed relative to the two individual input forecasts. Looking first at the full dataset results, the ML Ridge model achieved the best performance across the majority of metrics, recording the lowest MAE of 101.85 MW, the lowest RMSE of 132.94 MW, and the lowest MAPE of 2.10%, as well as the highest correlation of 0.9859. The EirGrid forecast performed the weakest across most metrics, with an MAE of 123.19 MW and an RMSE of 164.30 MW, while the External Forecast sat between the two, performing reasonably well but still falling short of the blended model.

Forecast Accuracy Comparison - ML Ridge (Full Data)
Green = Best performer for metric

Forecast	ME (Bias)	MAE	RMSE	MAPE (%)	Std Error	Max Error	Correlation
Forecast	-4.85	109.47	141.90	2.27%	141.83	1299.53	0.9843
EirGrid	-46.74	123.19	163.65	2.59%	156.84	1417.83	0.9815
ML Ridge	0.56	102.01	133.17	2.12%	133.17	1322.02	0.9861

Figure 20: Forecast accuracy comparison across all data (full dataset).

Forecast Accuracy Comparison - ML Ridge (Test Data Only)
Green = Best performer for metric

Forecast	ME (Bias)	MAE	RMSE	MAPE (%)	Std Error	Max Error	Correlation
Forecast	-3.05	109.67	142.55	2.27%	142.54	1282.33	0.9842
EirGrid	-43.84	123.19	164.30	2.57%	158.37	1413.67	0.9812
ML Ridge	2.78	102.66	134.07	2.13%	134.06	1309.73	0.9859

Figure 21: Forecast accuracy comparison on unseen test data only.

The test data results further reinforced these findings. On unseen data, the ML Ridge model again achieved the best MAE of 102.66 MW, the best RMSE of 134.07 MW, and the best MAPE of 2.13%, along with the highest correlation of 0.9859. Notably, the External Forecast achieved the lowest maximum error of 1282.33 MW on the test set, suggesting it handled extreme demand events slightly better in isolation. However, the ML Ridge model still outperformed both individual forecasts on average error metrics, demonstrating that the weighted blend was able to consistently produce more accurate predictions across the full range of demand conditions. The near-identical performance between the full dataset and test-only results further confirmed that the model had not overfit and was generalising effectively to new data.

In terms of absolute performance, an RMSE of approximately 134 MW and MAE of approximately 103 MW represented a meaningful improvement over using either individual forecast in isolation. The blended ensemble model was therefore able to produce a more stable and accurate prediction of system-wide power demand by combining the complementary strengths of both input forecasts.

Figure 22 illustrates the ML Ridge blended forecast against the actual outturn demand over a sample period. The black line represents the actual outturn, the green line the EirGrid forecast, and the red dashed line the ML Ridge blended forecast. The blended model tracks the outturn more closely than the EirGrid forecast throughout, with the middle panel confirming that the ML Ridge errors remain tightly centred around zero.

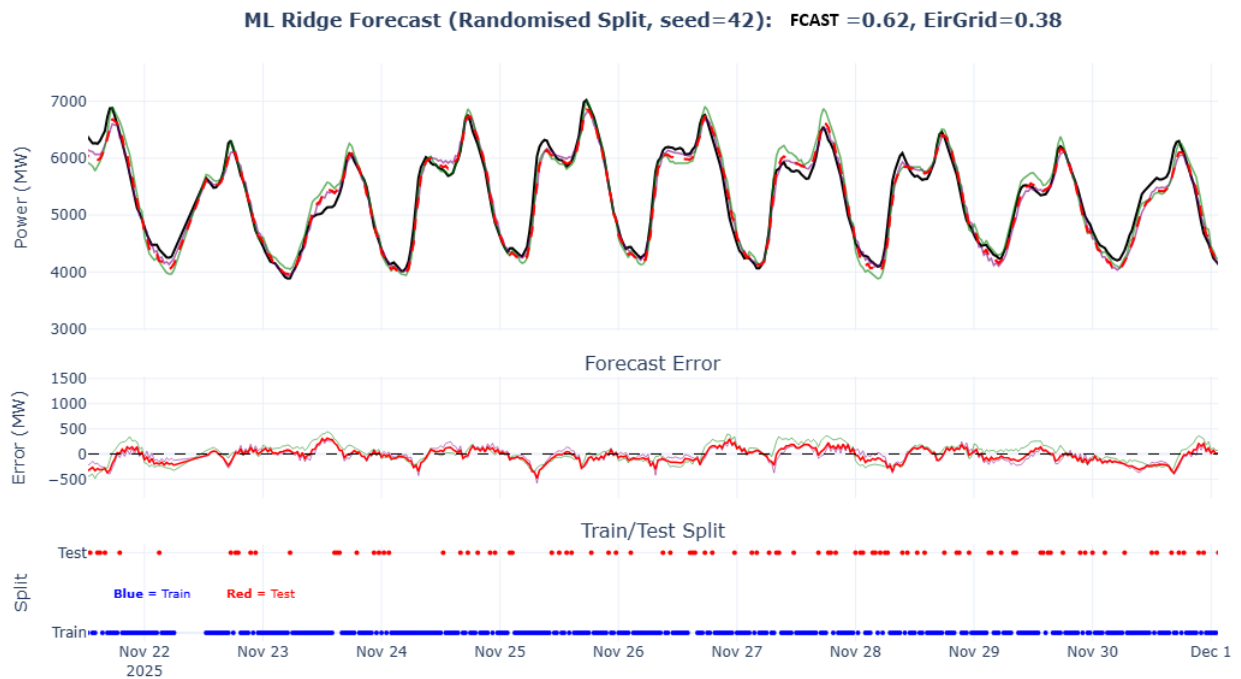


Figure 22: ML Ridge blended forecast vs. actual demand with error and train/test split.

The lower panel indicates the randomised train/test split, with blue and red dots representing training and test samples respectively.



Figure 23: Error distribution, cumulative error, scatter plot, and MAE/RMSE comparison.

5.4 Next Steps

While the ML Ridge blended model demonstrated a meaningful improvement over the individual input forecasts, there are several avenues through which the model could be further developed and refined.

The most immediate opportunity would be to incorporate additional input features beyond the two existing forecasts. Rooftop solar generation is a particularly relevant candidate, as it directly suppresses system demand during daylight hours and is not always fully captured in traditional demand forecasts. As rooftop solar capacity continues to grow across Ireland, its impact on outturn demand is likely to become increasingly significant, and failing to account for it could introduce a systematic bias into any demand forecast. Integrating rooftop solar generation estimates as an additional input feature could therefore improve forecast accuracy, particularly during summer months and periods of high irradiance. Beyond solar, other variables such as temperature, wind generation, and public holiday indicators could also be explored as additional inputs, given their known influence on electricity demand patterns. Extending the model to incorporate these features would likely require moving beyond ridge regression towards a more flexible modelling framework, such as gradient boosting or a neural network, which could better capture the non-linear relationships between demand and its drivers.

Finally, the model could be further developed to produce probabilistic forecasts rather than a single point estimate, providing the trading team with a measure of forecast uncertainty that could be directly incorporated into trading decisions. The model will continue to be worked on over the coming weeks, and hopefully by the end of my internship I will have contributed another valuable tool to the desk.

5.5 Skills and Learning Outcomes

Working with `scikit-learn` provided hands-on experience implementing machine learning pipelines in Python, including data preparation, model fitting, and performance evaluation. Applying ridge regression in a practical context developed a stronger understanding of how machine learning techniques can be applied to real-world forecasting problems. The project also reinforced broader statistical concepts around model validation and the interpretation of error metrics. Finally, analysing the drivers of forecast error developed a greater appreciation of the physical factors that influence electricity demand, including the growing impact of rooftop solar generation on system demand outturn.

6 Paper Trading and Trading Strategy

As part of the internship, paper trading was carried out on the REFIT asset portfolio in the Day-Ahead Market. Paper trading involves submitting simulated trades against live market data without actual financial exposure — the traded volumes do not enter the real auction and therefore do not influence clearing prices or volumes. While this removes the market impact element of real trading, it provides a valuable environment for developing trading intuition, testing strategies, and building a structured decision-making process against real market outcomes.

Over the course of the internship, a trading strategy framework was developed and iteratively refined based on the daily feedback loop that paper trading provided. Each morning, the available generation forecasts, weather data, system fundamentals, and market conditions were reviewed. A position was constructed using the framework described in this section, and after the auction cleared, the result was evaluated to identify what worked, what did not, and why.

6.1 Underselling and Overselling

The most fundamental decision in renewable energy trading is how much of the forecast generation to sell at the Day-Ahead auction versus holding volume for later markets. This is framed in terms of underselling and overselling relative to the generation forecast.

Underselling — selling less than the forecast — is typically motivated by:

- **Forecast uncertainty:** selling 100% of an unreliable forecast risks being short and paying high imbalance prices to buy back the shortfall.
- **Asymmetric imbalance penalties:** if the cost of being short exceeds the cost of being long, underselling provides a buffer.
- **Price expectations:** if IDA or imbalance prices are expected to exceed DA prices, holding back volume allows it to be sold later at better prices.
- **Flexibility:** selling 80% at DA preserves optionality to adjust in intraday auctions as forecasts improve.

Overselling — selling more than the forecast — is appropriate when:

- **Conservative forecasts:** if forecast providers consistently under-predict generation, selling above the forecast captures the expected upside.
- **High DA prices:** if DA prices are elevated and expected to fall, locking in revenue early is preferable.
- **System long conditions:** if the grid is expected to be long and imbalance prices low, overselling and buying back cheaply in the Balancing Market can be profitable.

6.2 Weather-Driven Factors

Unlike conventional commodities, renewable energy generation is fundamentally weather-dependent. Ireland's position at the western edge of Europe exposes it to Atlantic weather systems that can change conditions dramatically within hours. The following factors were monitored daily as part of the paper trading workflow.

Wind Speed

Wind speed is the most critical variable for renewable generation in Ireland given the high penetration of wind power on the grid. Table 5 summarises the trading implications across different conditions.

Condition	Impact	Strategy
Low (<5 m/s)	Under-generation, high prices	Undersell at DA.
Moderate (8–15 m/s)	Good generation, stable	Sell confidently at DA.
High (15–25 m/s)	Full generation, prices often low/negative	Sell at DA, avoid imbalance.
Storm (>25 m/s)	Cut-out risk: 100% to 0% output	Sell very conservatively.
Atlantic fronts	Rapid changes in generation	Hold for IDA1; DA forecasts unreliable.

Table 5: Wind speed conditions and corresponding trading strategies.

Wind turbines operate within specific thresholds: below the cut-in speed (typically 3–3.5 m/s) turbines do not generate, and above the cut-out speed (typically 25 m/s) turbines shut down for safety. Trading near these

thresholds requires particular caution, as a small forecast error can mean the difference between full output and zero generation.

Temperature, Cloud Cover, and Precipitation

Temperature is a primary driver of electricity demand in Ireland. Unlike continental Europe, Irish demand is predominantly driven by heating rather than cooling, making cold weather the key factor for price increases. A cold snap drives high demand and higher prices, favouring an aggressive DA sell, while mild or warm conditions suppress demand and reduce the urgency to sell at DA.

Cloud cover directly affects solar generation output. Ireland is typically overcast, meaning solar output is generally low. Clear skies and overcast conditions are both relatively predictable and favour selling at DA, whereas broken cloud introduces variability that may warrant holding volume for IDA. Global Horizontal Irradiance (GHI) correlates directly with solar panel output — high GHI ($>800 \text{ W/m}^2$) favours DA selling, while variable GHI favours holding for IDA.

Rain and fog reduce solar irradiance, while heavy precipitation events are often associated with changing wind patterns. Atmospheric pressure serves as a useful indicator of weather stability: high pressure brings settled conditions favouring confident DA selling, while low or falling pressure signals uncertainty and favours holding volume for IDA.

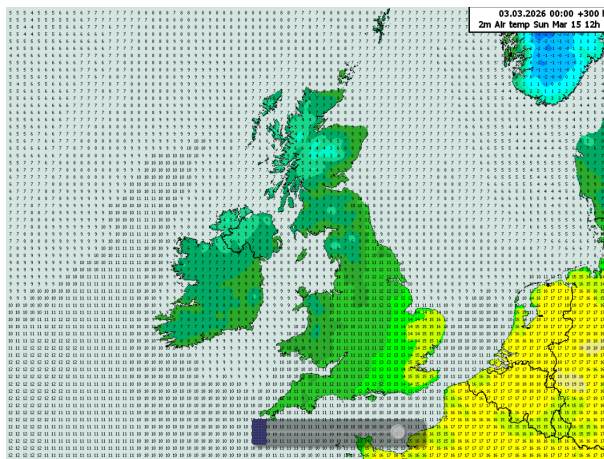


Figure 24: Forecasted temperatures.

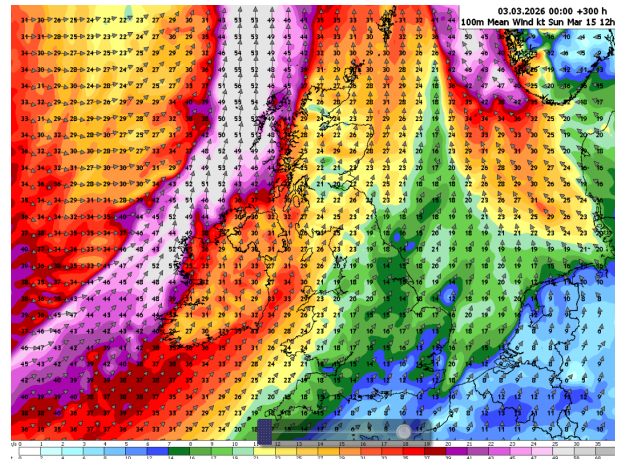


Figure 25: Forecasted wind speeds.

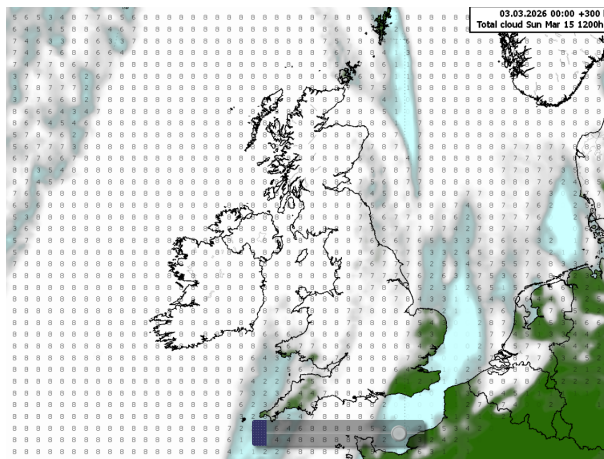


Figure 26: Forecasted cloud cover.

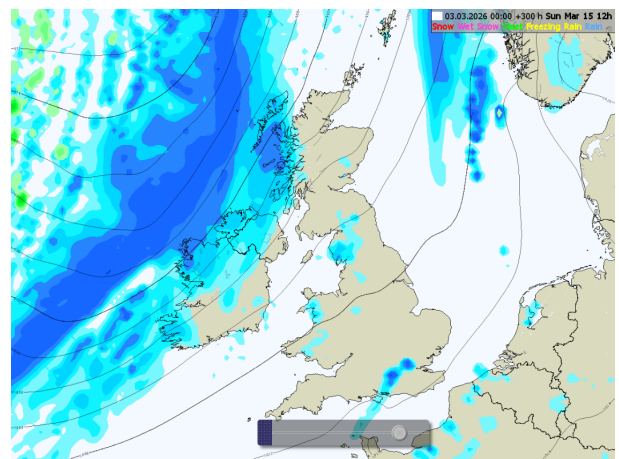


Figure 27: Forecasted precipitation.

6.3 System and Market Factors

Interconnector Flows

The EWIC and Moyle interconnectors link the I-SEM to the GB market. When Ireland is exporting to GB, I-SEM prices are compressed and selling at DA is straightforward. When importing from GB, I-SEM prices are higher, making DA volume more valuable. Constrained interconnectors introduce price volatility, favouring holding volume for IDA.

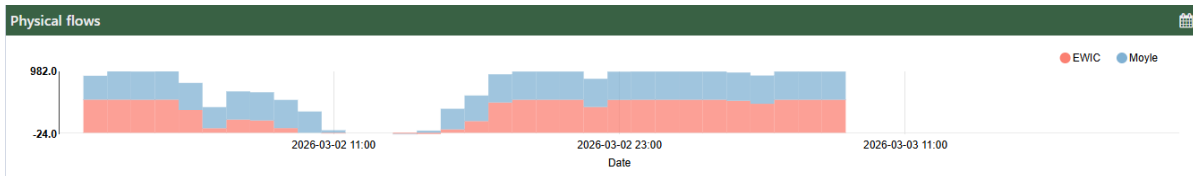


Figure 28: Power flows across the I-SEM/GB interconnectors. Positive values indicate an import.

Curtailment and SNSP

System Non-Synchronous Penetration (SNSP) measures the proportion of demand being met by non-synchronous generation. The current SNSP limit in Ireland is approximately 75%. Below 60%, curtailment is unlikely and volume can be sold normally. Between 60–70%, curtailment becomes possible and the TSO dispatch should be monitored. Above 75%, curtailment is highly likely and generators are not compensated for constrained volume, requiring a significant undersell.

Net Imbalance Volume and System Position

The Net Imbalance Volume (NIV) indicates whether the system as a whole is over-generating (long) or under-generating (short) relative to scheduled positions. The key principle is to be on the opposite side of the system: being long when the system is short means receiving a high imbalance price for excess generation, while being short when the system is long means paying a low imbalance price to cover the shortfall. Conversely, being on the same side as the system is unfavourable — being short when the system is short means paying a high imbalance price, and being long when the system is long means receiving a low price for excess generation.

This translates directly into strategy: if the system is expected to be long (e.g. high wind, low demand), underselling avoids being on the same side. If the system is expected to be short, overselling avoids expensive imbalance exposure. In uncertain conditions, selling to forecast minimises risk. An internal tool was used to estimate the probability of the system being long or short and the likely imbalance price, which informed the daily trading position.

6.4 System-Wide Forecasts

Demand Forecasts

System-wide demand forecasts from multiple providers were reviewed daily. The analytical approach involved three steps: first, comparing forecasts to normal demand levels to anticipate whether the system would be long or short; second, comparing DA and ID forecast revisions to gauge whether prices were likely rising or falling; and third, checking agreement between forecast providers (External and EirGrid) to assess confidence. When providers agreed, positions were taken with conviction. When they diverged, volume was held for IDA until clarity emerged.

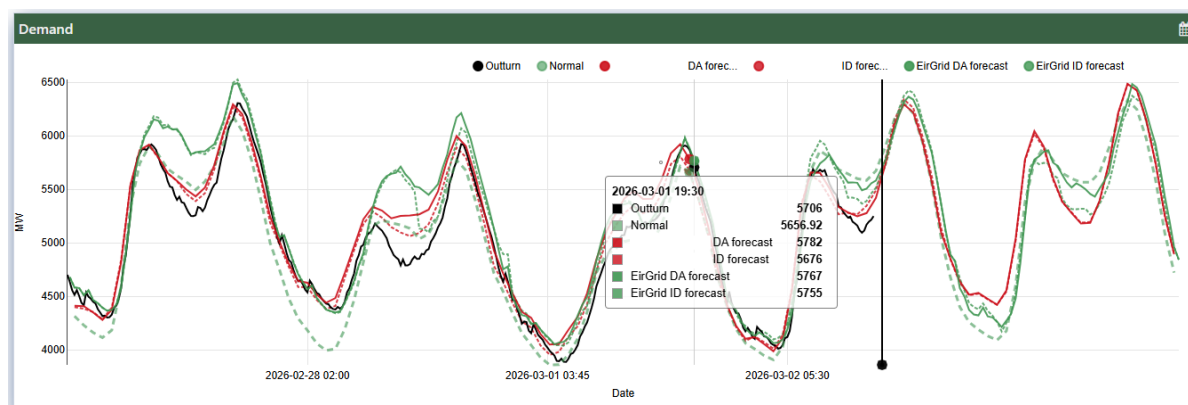


Figure 29: Actual and forecasted system demand from multiple providers.

Wind Generation Forecasts

Wind generation forecasts were analysed using a similar framework. The wind level was checked against SNSP to assess curtailment risk — high wind (>4000 MW) with high SNSP (>70%) signals a very long system requiring a significant undersell. The true (uncurtailed) wind was compared to the outturn forecast to identify expected curtailment. DA and ID forecast revisions were compared: if the ID forecast showed more wind than DA, prices were likely falling and the DA sell was well-timed; if less wind, prices were rising and holding for IDA was beneficial.

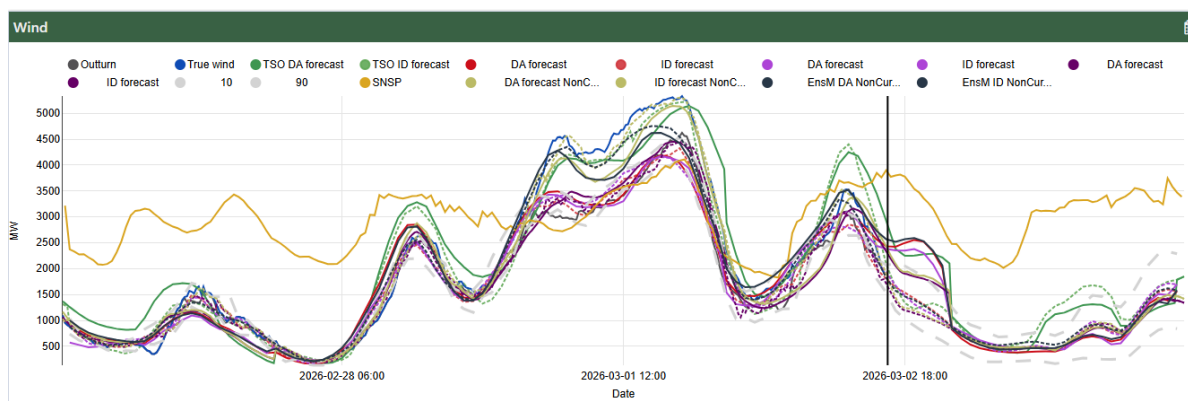


Figure 30: System-wide actual and forecasted wind generation.

Solar Generation Forecasts

Solar forecasts were assessed with particular attention to the P10/P90 spread as a measure of forecast confidence. A narrow spread (<20% of the median) supported selling 95% at DA, while a wide spread (>40%) suggested selling only 70%. Forecast provider agreement was checked — when all providers were within 10%, selling confidently was appropriate; when forecasts diverged, volume was held for IDA. Historical bias was also tracked: if a provider had been consistently over- or under-forecasting in recent conditions, the other providers were weighted more heavily.

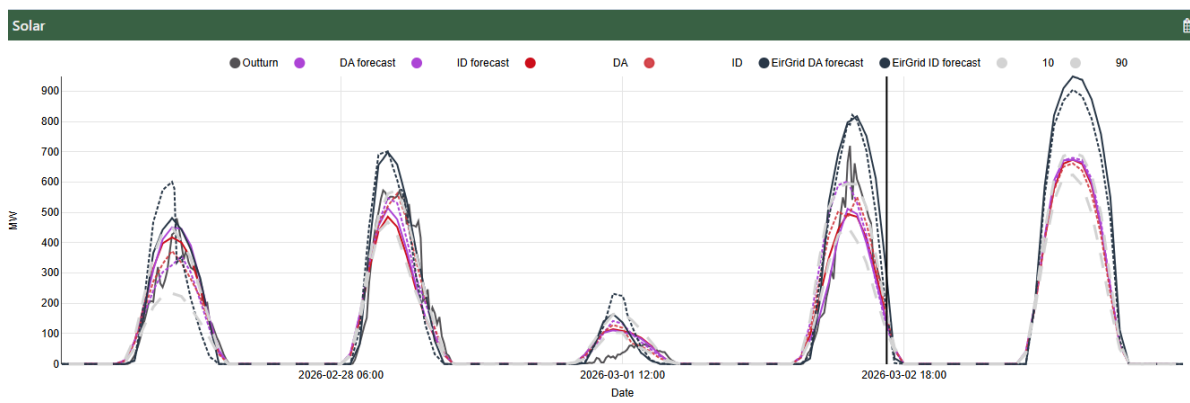


Figure 31: System-wide actual and forecasted solar generation.

6.5 Combined Assessment

In practice, no single factor determines the trading position. The variables described in the preceding subsections — weather conditions, system fundamentals, forecast confidence, interconnector flows, SNSP, NIV probability, and gas prices — all interact in complex ways, and their relative importance shifts depending on the specific conditions of the day. A high wind forecast, for example, might normally favour selling confidently at DA, but if SNSP is already near its limit and curtailment is likely, the same high wind forecast could warrant a significant undersell. Similarly, a cold snap driving high demand might suggest strong DA prices, but if interconnector imports are flooding the market, the price signal may be muted.

The framework outlined in this section is therefore best understood as a set of guiding principles rather than a deterministic decision tool. It provides a structured way to think through the key variables and their likely interactions, but the final position always requires judgement — weighing conflicting signals, assessing which factors are dominant on a given day, and accepting the inherent uncertainty in the decision. A decision matrix was used during the morning trading workflow to map out each factor and its current status, but its purpose was to ensure nothing was overlooked rather than to mechanically produce an answer.

Additional factors monitored daily included gas prices (particularly the NBP benchmark, as thermal units typically set the clearing price), planned outages of portfolio assets or interconnectors, day-of-week and seasonal demand patterns, and time-of-day effects such as the morning demand ramp, evening peak, and midday solar price suppression.

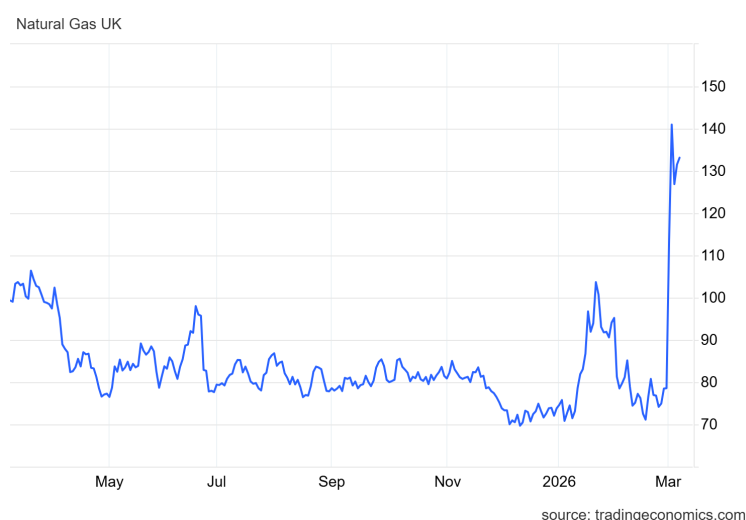


Figure 32: UK NBP Natural Gas futures prices as of 03/03/2025.

6.6 Paper Trading Results and Key Learnings

The results of each paper trading day were tracked and visualised using a custom plotting tool. Figure 33 shows an example result, with the upper panel displaying cumulative balancing costs over the trading day for both the paper-traded position and the Day-Ahead forecast baseline, and the lower panel showing the period-by-period breakdown of the difference.

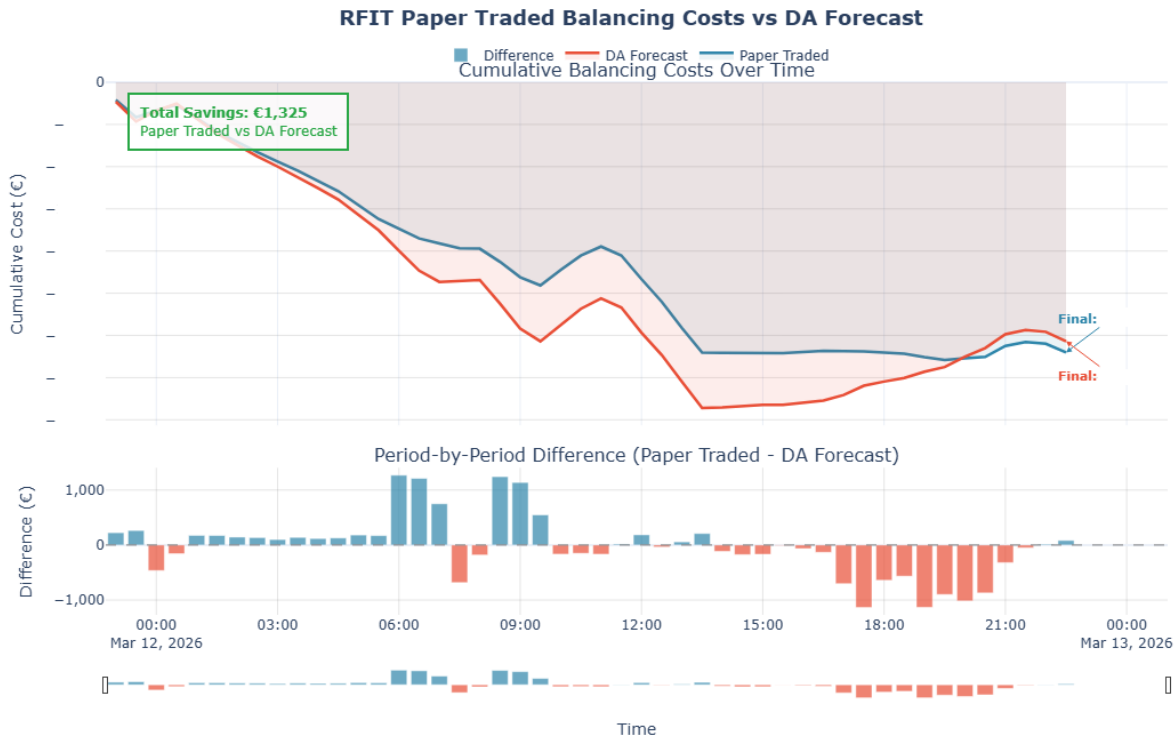


Figure 33: Example paper trading result for the REFIT portfolio, comparing cumulative balancing costs between the paper-traded position and the DA forecast baseline.

The paper trading exercise, combined with the iterative development of the trading strategy framework, reinforced several important lessons that would have been difficult to learn through analysis alone.

The most significant was developing an intuition for how forecast uncertainty translates into financial risk. Reviewing generation forecasts each morning and committing to a position forced a direct engagement with the question of how much to trust a forecast and how to hedge against the possibility of it being wrong. Over time, this built a practical understanding of the relationship between forecast confidence, traded volume, and exposure to imbalance prices.

The exercise also highlighted the importance of reading system fundamentals holistically rather than in isolation. On days with high wind penetration and low demand, for example, DA prices tended to be suppressed and the risk profile of holding volume into the Balancing Market changed accordingly. Learning to synthesise multiple signals — wind, demand, SNSP, interconnector flows, NIV probability — into a single coherent position was a skill that developed gradually through repetition.

Perhaps most importantly, the daily feedback loop provided a structured mechanism for learning from mistakes. Each trading day produced a clear, quantitative result that could be reviewed and reflected upon. This made it possible to identify patterns in decision-making — periods where positions were consistently too aggressive or too conservative — and to adjust the framework accordingly.

Although paper trading lacks the psychological pressure and market impact of live trading, it provided an essential foundation in the mechanics of energy trading and the discipline of structured decision-making under uncertainty. The trading strategy framework developed through this process represents a tangible, reusable output that makes use of the market knowledge accumulated over the course of the internship. Hopefully one day I will have the opportunity to swap the paper book for a real portfolio and become an energy trader.

7 Certifications and Professional Development

In addition to the project work described in this report, the internship included the completion of several professional development courses and certifications relevant to data science, energy markets, and regulatory compliance.

7.1 Dataquest: Data Scientist in Python

To strengthen the programming and data science skills required for the internship projects, the Dataquest *Data Scientist in Python* learning path was undertaken. This is a structured programme covering Python programming, data analysis, visualisation, data cleaning, SQL, machine learning, and advanced topics in data science. The complete learning path is available at <https://app.dataquest.io/learning/path/data-scientist/v2-7>. The courses completed to date are listed in Table 6.

Part	Course	Date Completed
1	Python Introduction	January 2025
2	Data Analysis and Visualisation	February 2025
3	Data Cleaning	April 2025
4	The Command Line	April 2025
5	Working with Data Sources Using SQL	In Progress
6	APIs and Web Scraping in Python	In Progress
7	Probability and Statistics	—
8	Machine Learning in Python	—
9	Deep Learning in Python	—
10	Advanced Topics in Data Science	—

Table 6: Dataquest *Data Scientist in Python* learning path progress. Completed course names link to their respective certificates.

The full *Data Scientist in Python* certification is pending completion of the remaining modules, which will be continued over the coming months alongside the extended internship contract.

7.2 Energy Markets Training

The *Energy Market Fundamentals* course offered by Energy Traders Europe is currently in progress. This course covers the structure and operation of European energy markets, including wholesale electricity and gas trading, market coupling, and the role of exchanges and clearing houses. Completing this course will

complement the practical market knowledge gained through the internship with a broader understanding of European energy market frameworks.

7.3 Regulatory Compliance

The *Anti-Bribery Fundamentals* training course was completed as part of ElectroRoute's mandatory compliance programme. This course covered the legal and ethical obligations relating to anti-bribery and corruption in a commercial trading environment, including the identification of bribery risks, reporting obligations, and the company's internal policies and procedures.

8 Conclusion

This internship has been the most formative experience of my academic and professional career to date. Working within ElectroRoute's Trading Operations team for five months provided an immersion into the commercial realities of renewable energy trading that no classroom setting could replicate. The opportunity to work on projects with genuine business impact — tools and analysis that the team will continue to use after I leave — gave the work a sense of purpose and accountability that pushed me to produce my best work.

From a technical perspective, the placement transformed my ability to work with data. At the start of the internship, my experience with Python and SQL was largely self-motivated, with prior academic experience limited to C++ and MATLAB. By the end, I was writing production-level code, building full-stack applications, and applying machine learning techniques to real forecasting problems. Learning to work with large, messy, real-world datasets, and learning how to extract meaningful insight from them, was a fundamentally different challenge to anything I had encountered in university, and one that I found deeply rewarding.

Equally valuable was the commercial and market knowledge I developed over the course of the placement. Understanding how the I-SEM operates, how renewable generators are traded, and how trading decisions are made under uncertainty gave me a perspective on the energy system that extends well beyond the engineering fundamentals covered in my degree. Sitting on the trading desk, attending market calls, and working alongside the traders allowed me to ask questions and learn from some of the most experienced individuals in the industry.

The experience also reinforced the importance of communication. Whether presenting my dashboard to the CEO or delivering analysis to the trading team, the ability to convey technical work clearly and concisely proved just as important as the technical work itself. This is something I will carry forward into every professional setting.

Looking ahead, this internship has firmly inspired me to pursue a career in energy trading once I graduate from my Masters in Engineering next year. The combination of quantitative problem-solving, real-time decision-making, and direct engagement with energy markets is something I find genuinely exciting. I leave this placement with a strong foundation in the skills and knowledge required to contribute in this space, and a clear ambition to build a career at the intersection of energy markets, data science, and trading.

For my final year, I would like to complete my Masters Research Project in an area that bridges energy markets and energy systems engineering. The energy sector is undergoing a period of rapid transformation, driven by increasing global demand, the integration of variable renewable generation, and the electrification of transport and heating. What was once considered a slow-moving industry has become one of the most dynamic, technology-driven spaces globally, creating new opportunities across markets, infrastructure, and innovation. It is an exciting time to be entering the field.

I am deeply grateful to the team at ElectroRoute for the opportunity, the trust, and the mentorship they provided throughout the year. While the internship report covers the period up to the end of May, my contract extends through to August. In the months ahead, I hope to continue contributing to the team and

deepening my understanding of the energy industry, building a foundation that I can carry into a graduate role.